

Structure-preserving physics-informed learning of open quantum dynamics with spectral-frame encodings and operator-system truncation

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Characterizing a quantum device is fundamentally a problem of statistical inference: the hardware is only ever interrogated by sampling, so identifying its dissipative dynamics means fitting a probabilistic surrogate to finite, noisy measurements and quantifying what that surrogate can and cannot certify. We treat open-system identification in exactly this way and introduce a structure-preserving, physics-informed framework for identifying Markovian open quantum systems through three architectural innovations that make the standard failure modes of neural surrogates unrepresentable by construction. First, a Cholesky density network maps every time point to a Hermitian, positive-semidefinite, unit-trace state, so no training step can emit an unphysical output. Second, a spectral-frame (interaction-picture) parameterization learns only the slow envelope in the Hamiltonian eigenbasis and supplies the fast Bohr-frequency oscillations analytically, removing spectral bias at the architecture level. Third, an operator-system spectral truncation band-limits the generator to its slow Bohr modes, yielding a controlled hierarchy from the secular (Davies) generator through the rotating-wave approximation to the exact generator with an a posteriori error certificate. We prove eight exact guarantees, including completely-positive trace-preserving structure and the truncation bound. On reproducible simulations the spectral frame raises fast qutrit (leakage) reconstruction from mean fidelity 0.5–0.8, with confidence intervals as wide as the means, to 0.985 ± 0.002 ; the hard map holds the trace distance one to two orders of magnitude below soft-penalty and unconstrained baselines under sparse data; and the learned two-qubit generator transfers to held-out initial states at ≈ 0.97 mean fidelity. The study is fully reproducible and CPU-scale rather than a hardware benchmark, delivering a mathematically constrained, spectrally structured, accelerator-aware formulation of open-system identification.

I. INTRODUCTION

Statistics is under-rated in, and structural to, quantum computing. From the outside a quantum processor is a stochastic device: the Born rule means every run returns a single bitstring drawn from a distribution, so everything one learns about the machine—estimating an observable, reconstructing a state, certifying a gate, tracking calibration drift—is an act of statistical estimation and inference from samples. Characterizing the dissipative dynamics of the hardware is no exception. It is an inverse problem in which an expensive physical experiment is emulated by a learned surrogate that must predict trajectories, and the noise processes that generate them, from finitely many noisy observations and with calibrated uncertainty about what the data support.

Open quantum systems are the operational language of quantum hardware. A superconducting, trapped-ion, photonic, spin, or neutral-atom processor is never described by its ideal Hamiltonian alone; relaxation, dephasing, leakage, crosstalk, drift, and control-induced dissipation are equally part of its description, and characterizing them from data is a prerequisite for calibration and control. The object that must remain physical throughout any such description is the density matrix, which for a finite Hilbert space $\mathcal{H} \cong \mathbb{C}^d$ ranges over the state space

$$\mathcal{D}(\mathcal{H}) = \{\rho \in \mathbb{C}^{d \times d} : \rho = \rho^\dagger, \rho \succeq 0, \text{Tr } \rho = 1\}. \quad (1)$$

When the dynamics are Markovian and time-local, the most general generator of a completely positive, trace-preserving (CPTP) semigroup is the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) form [1–3], the backbone of quantum noise modeling, control, and calibration [4–7].

Inferring such a generator from incomplete observations—the inverse problem of open-system identification—is increasingly central to device characterization. Classical process tomography reconstructs a channel but scales poorly with system size [8–10], while Hamiltonian- and Lindblad-learning exploit dynamical structure yet still face identifiability and optimization obstacles [11, 12]. Physics-informed neural networks (PINNs) supply flexible function approximation with the equation of motion placed directly in the training objective [13–15], but a direct neural map $t \mapsto \rho_\theta(t)$ has no reason to remain Hermitian, positive semidefinite, or unit trace. Penalizing negative eigenvalues is not equivalent to physicality during training; it merely asks the optimizer to repair an invalid state after the fact.

We take the opposite design principle: *make invalid states unrepresentable*. The density network is

$$\rho_\phi(t) = \frac{A_\phi(t)A_\phi(t)^\dagger}{\text{Tr}[A_\phi(t)A_\phi(t)^\dagger]}, \quad (2)$$

where $A_\phi(t)$ is a neural lower-triangular complex matrix with nonzero diagonal. This Cholesky-type map is differentiable and physical at the architecture level: it cannot represent a state outside $\mathcal{D}(\mathcal{H})$. Dissipation rates are pa-

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parameterized by softplus transforms and full Kossakowski matrices as Gram matrices $BB^\dagger$, so the learned generator never leaves GKSL form.

A hard physicality constraint does not by itself address a second, orthogonal obstacle. A direct map $t \mapsto \rho_\phi(t)$ must represent every fast coherent oscillation in the lab-frame trajectory, and a smooth-in-time network suffers *spectral bias*: it learns low frequencies first and resolves the rapid Bohr-frequency oscillations of multilevel systems only slowly and unreliably [44]. We remove this obstacle at the architecture level with a *spectral-frame* (interaction-picture) parameterization. The coherent generator $\text{ad}_H = -i[H, \cdot]$ is normal; in the Hamiltonian eigenbasis its eigen-operators are the transition operators $|a\rangle\langle b|$ with purely imaginary eigenvalues $-i\omega_{ab}$, $\omega_{ab} = E_a - E_b$ (the Bohr frequencies), which are precisely the fast oscillations. The network therefore learns only the *slow envelope* $\tilde{\rho}(t) = U(t)^\dagger \rho(t) U(t)$, $U(t) = e^{-iHt}$, through the *same* Cholesky map—so the envelope, and hence $\rho(t) = U(t)\tilde{\rho}(t)U(t)^\dagger$, is physical by construction—while the fast oscillation is supplied analytically by the known $U(t)$. Because conjugation by a unitary preserves $\mathcal{D}(\mathcal{H})$, physicality remains architectural, and the trainable target is slow, so spectral bias is removed rather than penalized.

The same eigen-operator decomposition turns the residual kernel itself into an object of *spectral truncation*. In the interaction picture the generator $\tilde{\mathcal{L}}_t$ is almost periodic; on a finite window it has a Fourier series $\tilde{\mathcal{L}}_t = \sum_k e^{ik\omega_0 t} \mathcal{L}_k$, and the level- N truncation keeps the low-frequency band $|k| \leq N$ —a projection onto a finite *operator system* of slow eigen-operators. This is the spectral-truncation construction by which noncommutative geometry approximates a C^* -algebra by finite-dimensional operator systems [40, 41] and by which C^* -algebraic kernel machines reinstate noncommutativity [42]; here we lift it from static kernels to a *dynamical, completely positive generator*. The truncation is a controlled, multi-resolution hierarchy—the $N = 0$ (secular) level is the time-independent Davies generator [43], $N = 1$ the rotating-wave approximation, and the full band the exact generator—equipped with an a posteriori certificate (the out-of-band spectral weight).

We use the term *compiler* in a precise sense. A quantum model is specified in an intermediate representation (IR): Hilbert dimension, Hamiltonian terms, jump operators, rates, observables, batching rules, and differentiation mode. The compiler lowers this IR to a dense Liouvillian, a structured residual kernel, or a spectrally truncated (operator-system) kernel. For small d a dense kernel is convenient; for larger d the structured route stores H and m Lindblad operators rather than a $d^2 \times d^2$ matrix, trading $O(d^4)$ storage for $O(md^2)$. This mirrors the design philosophy of contemporary accelerator software—represent intent at a high level, then lower to hardware-conscious kernels—and is naturally compatible with PyTorch [16], SciPy [17], open-system toolkits such as QuTiP [18], circuit frameworks such as Qiskit [19],

GPU-accelerated backends, and future custom kernels.

Relation to prior work. The method follows the structured-identification philosophy of Lindblad and Hamiltonian learning [11, 12] but replaces unconstrained trajectory estimation with a hard density-matrix parameterization, in contrast to completely-positive *projection* methods that repair an estimate only after the fact [10]. It is closest to inverse PINNs [13, 14, 20], with which it shares the residual objective, and to structure-preserving scientific networks—Hamiltonian, Lagrangian, symplectic—that restrict the hypothesis class to improve learning [21–23]. The spectral-frame parameterization adopts the interaction-picture (eigen-operator) decomposition of open-system dynamics [3, 43] as an *architectural* device, complementary to deterministic Fourier-feature encodings used to counter spectral bias [44]. Its operator-system spectral truncation of the generator transfers, for the first time, the spectral-truncation construction of noncommutative geometry [40, 41]—recently used to build noncommutative C^* -algebraic kernel machines [42]—from *static kernels* to the *learning and approximation of a completely positive dynamical generator*, with the secular/Davies generator as its coarsest level. It is complementary to neural quantum states and neural tomography, which represent wavefunctions, density operators, or measurement distributions with flexible models [24–26], and to operator-learning models that target solution *families* rather than a single trajectory [15, 27, 28]. We begin with the Markovian case; non-Markovian systems require extended state spaces, memory kernels, or process tensors [29–32], and dissipation can itself be a computational resource [33]. This paper is the open-system entry in a single-author program on spectral truncation of operators, which treats query-efficient and topology-conditioned QAOA [45–51], scheduled and Lie-algebraic product-formula error control [52, 53], and learned decoding of structured quantum noise [54–57], as instances of one construction: approximate a high-dimensional operator by a controlled, certifiable projection onto a slow or low-weight sub-band. It inherits in particular the resource-accounted discipline of graph- and topology-conditioned QAOA [45, 51] and commutator-graph Hamiltonian simulation [52], and the residual-learning perspective developed for Trotter corrections as a closed-system counterpart [53]; the Hamiltonian-simulation work of Nguyen and Jing supplies further fixed-depth and representation-theoretic context [58, 59], while the present contribution moves to open-system, CPTP-constrained identification rather than product-formula simulation.

In this work, we introduce four constructions and prove eight guarantees about them. *First*, a hard-constrained density network whose Cholesky map [Eq. (2)] enforces Hermiticity, positive semidefiniteness, and unit trace exactly at every time point, every iteration, and every evaluation point. *Second*, a spectral-frame eigen-operator parameterization that learns the slow interaction-picture envelope through the same hard map and supplies the

fast Bohr oscillations analytically; this resolves the principal open problem of the structure-preserving approach—reliable reconstruction of fast multilevel (qutrit leakage) dynamics across random seeds (mean state fidelity 0.985 ± 0.002)—and, by conditioning the envelope derivative, renders the dissipative rates identifiable by a linear least-squares readout (dominant-rate relative error 0.273 ± 0.025). *Third*, an operator-system spectral truncation of the generator that band-limits it to its slow Bohr eigen-operators, giving a controlled hierarchy from the secular (Davies) generator to the exact generator with an a posteriori out-of-band certificate (Theorem 8) and lifting the spectral-truncation construction of noncommutative geometry [40, 42] to a completely positive dynamical generator. *Fourth*, a CPTP generator parameterization with a matrix-free compiler that lowers a symbolic open-system IR to a dense superoperator, a structured residual kernel, or a spectrally truncated kernel while preserving GKSL form (Theorem 6). Every figure and number is produced by the accompanying `ctpcpinn` package; each training experiment is repeated over five seeds and reported as mean $\pm$ a 95% confidence interval, against a soft positivity penalty, an unconstrained network, plain and Fourier-feature density networks, and classical least-squares fits.

These results establish what the structural constraints, the spectral frame, and the compiler path deliver on controlled, fully reproducible simulations. They are deliberately scoped to that setting: quantum-hardware validation, superiority over every baseline, and production-scale GPU performance remain future experimental targets rather than claims made here.

II. STRUCTURE-PRESERVING FRAMEWORK

Figure 1 gives an overview. A density network maps each time point to a physically valid state by construction; a separate, structure-preserving parameterization keeps the learned generator completely positive; and a matrix-free compiler evaluates the Lindblad residual that couples them to data.

A. Hard-constrained density parameterization

Let $B_\phi : [0, T] \rightarrow \mathbb{C}^{d \times d}$ be a neural network whose output is read as a lower-triangular matrix $A_\phi(t)$ with nonzero diagonal,

$$A_\phi(t)_{ij} = \begin{cases} 0, & i < j, \\ a_{\min} + \text{softplus}(b_{ii}(t)), & i = j, \\ b_{ij}(t), & i > j, \end{cases} \quad (3)$$

with $a_{\min} > 0$ a small numerical floor, and define $\rho_\phi(t)$ by Eq. (2). The output is d^2 real degrees of freedom; a full-rank density matrix has $d^2 - 1$ after trace normalization, so the representation is only mildly redundant, and

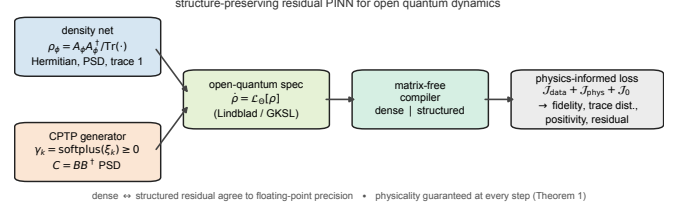


FIG. 1. The structure-preserving, physics-informed framework. Two structure-preserving networks feed a physics-informed residual. The density network maps time to a lower-triangular factor $A_\phi(t)$ and outputs $\rho_\phi = A_\phi A_\phi^\dagger / \text{Tr}(A_\phi A_\phi^\dagger)$, which is Hermitian, positive semidefinite, and unit trace at every time and every training step (Theorem 1); the generator is parameterized so that all Lindblad rates are nonnegative $[\gamma_k = \text{softplus}(\xi_k) \geq 0]$ or the Kossakowski matrix is positive semidefinite $[C = BB^\dagger]$, keeping $\mathcal{L}_\Theta$ in GKSL form (Theorem 3). A matrix-free compiler lowers the specification $\dot{\rho} = \mathcal{L}_\Theta[\rho]$ to a dense $d^2 \times d^2$ Liouvillian or a structured residual kernel (the two agree to floating-point precision; Theorem 6). Training minimizes a physics-informed loss combining the masked data term, the Lindblad residual, and the initial condition; the outputs are a physically valid trajectory, a completely positive generator, and the metrics (state fidelity, trace distance, positivity-violation rate, and the trace-norm residual certificate of Theorem 4). The figure is generated programmatically by `ctpcpinn/plotting.py:plot_schematic`.

rank-deficient (pure) states arise as limits. The map is differentiable and, by Theorem 1, always lands in $\mathcal{D}(\mathcal{H})$.

B. Spectral-frame eigen-operator parameterization

When the coherent dynamics are fast relative to dissipation, the same Cholesky map is applied to the *interaction-picture envelope* rather than the lab state. With frame Hamiltonian $H = \sum_a E_a |a\rangle\langle a|$ and $U(t) = e^{-iHt}$, the network outputs an envelope $\tilde{\rho}_\phi(t) \in \mathcal{D}(\mathcal{H})$ in the eigenbasis and the lab state is the analytic rotation $\rho_\phi(t) = U(t)\tilde{\rho}_\phi(t)U(t)^\dagger$ (a pure phase $e^{-i\omega_{ab}t}$ on the (a, b) entry when H is diagonal). By Theorem 7 this is exactly physical and exactly equivalent to the lab dynamics, but the trainable target $\tilde{\rho}_\phi$ is slow, so the network no longer has to resolve the fast Bohr-frequency oscillations: spectral bias is removed at the architecture level (Sec. V D).

C. Completely-positive generator parameterization

For known jump operators with unknown rates we write $\gamma_k(\xi_k) = \text{softplus}(\xi_k) + \gamma_{\min}$, $\gamma_{\min} \geq 0$; for a full Kossakowski matrix in a fixed operator basis $\{F_a\}$ we set $C_\eta = B_\eta B_\eta^\dagger \succeq 0$. Hamiltonians are parameterized in a Hermitian basis $H_\theta(t) = \sum_\ell \theta_\ell(t) G_\ell$. By Theorem 3 these choices keep the learned generator a GKSL generator (details in Appendix B).

D. Physics-informed objective and matrix-free compiler

The training objective combines a masked data term, a Lindblad physics residual evaluated at collocation points, and an initial-condition term,

$$\mathcal{J}(\phi, \Theta) = \lambda_{\text{data}} \mathcal{J}_{\text{data}} + \lambda_{\text{phys}} \mathcal{J}_{\text{phys}} + \lambda_0 \mathcal{J}_0, \quad (4)$$

with $\mathcal{J}_{\text{phys}} = \frac{1}{N_c} \sum_{\ell} \|\dot{\rho}_{\phi}(\tau_{\ell}) - \mathcal{L}_{\Theta}[\rho_{\phi}(\tau_{\ell})]\|_F^2$. We compute the time derivative $\dot{\rho}_{\phi}$ by automatic differentiation of the density network with respect to time; a centered finite-difference surrogate is an alternative when automatic differentiation is unavailable. The residual $\mathcal{L}_{\Theta}[\rho_{\phi}]$ is produced by a compiler that lowers the open-system IR to a dense Liouvillian or a structured matrix-free kernel, with a dispatch policy selecting the mode by dimension and batch size (Appendix F, Algorithms 1–2).

III. THEORETICAL GUARANTEES

The framework is supported by eight results, stated here and proved in Appendix G. They concern the architecture, the spectral frame, and the learned generator, and are exact rather than statistical.

Theorem 1 (Exact density-matrix physicality). *Let $A : [0, T] \rightarrow \mathbb{C}^{d \times d}$ be continuous with $A(t) \neq 0$ for all t , and let $\rho_A(t) = A(t)A(t)^{\dagger} / \text{Tr}(A(t)A(t)^{\dagger})$. Then $\rho_A(t) \in \mathcal{D}(\mathcal{H})$ for every t . If A is differentiable, ρ_A is differentiable, Hermitian, and $\text{Tr} \dot{\rho}_A(t) = 0$.*

Theorem 2 (Universal approximation of continuous density trajectories). *Let $\rho : [0, T] \rightarrow \mathcal{D}(\mathcal{H})$ be continuous and $\sigma \in \mathcal{D}(\mathcal{H})$ full rank. For every $\varepsilon > 0$ there is a finite network $A_{\phi} : [0, T] \rightarrow \mathbb{C}^{d \times d}$ with nonzero output such that $\sup_t \|A_{\phi} A_{\phi}^{\dagger} / \text{Tr}(A_{\phi} A_{\phi}^{\dagger}) - \rho(t)\|_F < \varepsilon$.*

Theorem 3 (GKSL preservation). *With $H_{\theta} = H_{\theta}^{\dagger}$ and $\gamma_k(\xi_k) = \text{softplus}(\xi_k) + \gamma_{\min}$, $\gamma_{\min} \geq 0$, the generator is a GKSL generator and $e^{t\mathcal{L}_{\Theta}}$ is CPTP for all $t \geq 0$. The Kossakowski parameterization with $C = BB^{\dagger}$ is likewise a GKSL generator.*

Theorem 4 (A posteriori residual certificate). *Let ρ solve $\dot{\rho} = \mathcal{L}[\rho]$, $\rho(0) = \rho_0$, with $\mathcal{L}$ generating a CPTP semigroup Φ_t , and let $\hat{\rho}$ be a differentiable Hermitian trace-one path with residual $r(t) = \dot{\hat{\rho}}(t) - \mathcal{L}[\hat{\rho}(t)]$. Then for all t ,*

$$\frac{1}{2} \|\hat{\rho}(t) - \rho(t)\|_1 \leq \frac{1}{2} \|\hat{\rho}(0) - \rho_0\|_1 + \frac{1}{2} \int_0^t \|r(s)\|_1 \, ds. \quad (5)$$

Theorem 5 (Local identifiability). *Let $\Theta \in \mathbb{R}^q$ parameterize a smooth GKSL generator with noiseless observation map $f_{ij}(\Theta) = \text{Tr}[O_j \rho_{\Theta}(t_i)]$ and Jacobian $J(\Theta_*) \in \mathbb{R}^{M \times q}$ at the true Θ_* . If $J(\Theta_*)$ has full column rank with smallest singular value $s_{\min} > 0$, then on a neighborhood*

where the Jacobian varies by at most $s_{\min}/2$, any estimate with $\|f(\hat{\Theta}) - f(\Theta_)\|_2 \leq \delta$ obeys $\|\hat{\Theta} - \Theta_*\|_2 \leq 2\delta/s_{\min}$.*

Theorem 6 (Dense versus structured residual complexity). *For a d -dimensional system with m dense Lindblad operators, dense Liouvillian storage is $O(d^4)$ and construction $O(md^4)$, with $O(d^4)$ per dense residual matvec; structured evaluation costs $O(md^3)$ arithmetic and $O(md^2)$ storage. For N collocation states, dense costs $O(md^4 + Nd^4)$ and structured $O(Nmd^3)$.*

Theorem 7 (Spectral-frame physicality and equivalence). *Let $U(t) = e^{-iHt}$ for a Hermitian frame Hamiltonian H , and let $\tilde{\rho} : [0, T] \rightarrow \mathcal{D}(\mathcal{H})$ be a (Cholesky-parameterized) envelope. Then $\rho(t) = U(t)\tilde{\rho}(t)U(t)^{\dagger} \in \mathcal{D}(\mathcal{H})$ for every t , so the spectral-frame map is exactly physical. Moreover ρ solves the lab master equation $\dot{\rho} = \mathcal{L}_{\Theta}[\rho]$ if and only if $\tilde{\rho}$ solves the interaction-picture equation $\dot{\tilde{\rho}} = \tilde{\mathcal{L}}_{\Theta,t}[\tilde{\rho}]$, whose generator $\tilde{\mathcal{L}}_{\Theta,t}[\cdot] = -i[\tilde{H}_c(t), \cdot] + \sum_k (\tilde{L}_k(t) \cdot \tilde{L}_k(t)^{\dagger} - \frac{1}{2}\{\tilde{L}_k(t)^{\dagger} \tilde{L}_k(t), \cdot\})$ is GKSL for each t , with $\tilde{L}_k(t) = U(t)^{\dagger} L_k U(t)$ and $\tilde{H}_c = U^{\dagger}(H_{\Theta} - H)U$.*

Theorem 8 (Operator-system spectral-truncation certificate). *On $[0, T]$ the interaction-picture generator has the Fourier series $\tilde{\mathcal{L}}_t = \sum_{k \in \mathbb{Z}} e^{ik\omega_0 t} \mathcal{L}_k$, $\omega_0 = 2\pi/T$, with $\sum_k \|\mathcal{L}_k\| < \infty$. Let $\tilde{\mathcal{L}}_t^{(N)} = \sum_{|k| \leq N} e^{ik\omega_0 t} \mathcal{L}_k$ be the level- N spectral truncation. Then $\sup_t \|\tilde{\mathcal{L}}_t - \tilde{\mathcal{L}}_t^{(N)}\| \leq \sum_{|k| > N} \|\mathcal{L}_k\| =: \eta_N \rightarrow 0$, and each $\tilde{\mathcal{L}}_t^{(N)}$ is Hermiticity- and trace-preserving (it retains $\mathcal{L}_0$ and conjugate-symmetric pairs $\mathcal{L}_{\pm k}$). Combined with Theorem 4, the trajectory error of the level- N truncated model is bounded by the initial error plus $\frac{1}{2} \int_0^t \eta_N \|\tilde{\rho}(s)\|_1 \, ds$, vanishing at full band ($N \rightarrow \infty$, recovering the structured/dense kernels exactly).*

Corollary 15 extends physicality to low-rank-plus-isotropic variants, and Proposition 16 gives a Frobenius analogue of Theorem 4 via Grönwall's inequality (Appendix G). Theorem 1 is operationally decisive: no training step can output a negative-probability state unless the floating-point implementation itself fails. Theorem 5 likewise explains why some inverse problems remain ill-conditioned even with hard constraints—if the measurement set does not excite a rate or Hamiltonian term, the sensitivity Gramian is singular.

IV. STATISTICAL AND APPROXIMATION THEORY OF THE STRUCTURE-PRESERVING SURROGATE

The guarantees of Sec. III are exact structural facts about the architecture. This section develops the complementary *statistical* and *approximation* theory that justifies the surrogate viewpoint: a physics-informed network is a probabilistic emulator of an expensive open-system

simulator, trained from finitely many noisy measurements and finitely many collocation points, whose predictions must therefore be quantified with error and uncertainty. We prove five further results. First, a quantitative bound on the completely-positive-trace-preserving (CPTP) violation of an *approximate* learned generator, showing that the $C = BB^\dagger$ parameterization sets that violation to exactly zero while a soft-penalized generator leaves a residual proportional to the negative part of its Kossakowski matrix (Proposition 9). Second, a sharpened a posteriori stability estimate whose constant is the logarithmic norm (dissipativity) of the Lindblad generator, so that on contractive dynamics the residual-to-error bound is *uniform in time* rather than exponentially growing (Lemma 10, Theorem 11). Third, an approximation-plus-generalization decomposition of the total surrogate error into an expressivity term (network approximation on the operator-system truncation) and an estimation term (finite collocation), yielding an explicit collocation-sample complexity through a uniform concentration of the empirical physics residual (Theorem 12). Fourth, a combination of that decomposition with the spectral-truncation tail η_N of Theorem 8 that identifies the cost-optimal truncation level (Corollary 13). Fifth, a Bayesian uncertainty-quantification statement: under the local-identifiability Gramian of Theorem 5, the surrogate posterior contracts at the parametric $n^{-1/2}$ rate and its push-forward to trajectory space yields asymptotically calibrated credible bands (Theorem 14). All proofs are in Appendix H; they reuse the CPTP/GKSL machinery, the trace-distance contraction, Grönwall’s inequality, and Rademacher/covering arguments, and make no assumption beyond those already in force.

A. Structure preservation of approximate generators

The exact GKSL preservation of Theorem 3 presumes the architectural map $C = BB^\dagger$. It is useful to know how far an *arbitrary* Hermitian Kossakowski matrix—for instance the one a soft-penalty method converges to—can be from generating a CPTP semigroup, and that the hard map annihilates exactly this quantity.

Proposition 9 (Quantitative CPTP violation and its exact removal). *Fix a Hermitian operator basis $\{F_a\}_{a=1}^q$ and a Hermitian Hamiltonian H , and for a Hermitian Kossakowski matrix $C \in \mathbb{C}^{q \times q}$ let $\mathcal{L}_C[\rho] = -i[H, \rho] + \sum_{a,b} C_{ab}(F_a \rho F_b^\dagger - \frac{1}{2}\{F_b^\dagger F_a, \rho\})$. Write $C = C_+ - C_-$ for its Jordan decomposition into positive and negative parts ($C_\pm \succeq 0$, $C_+ C_- = 0$). Then:*

1. $\{e^{t\mathcal{L}_C}\}_{t \geq 0}$ is CPTP for all $t \geq 0$ if and only if $C \succeq 0$, i.e. $C_- = 0$.
2. Let $\mathcal{L}_C^\sharp[\rho] = -i[H, \rho] + \sum_{a,b} (C_+)_{ab}(F_a \rho F_b^\dagger - \frac{1}{2}\{F_b^\dagger F_a, \rho\})$ be the CPTP projection obtained by discarding C_- . Then $\mathcal{L}_C^\sharp$ is a legal GKSL generator and,

in the induced superoperator norm on the Frobenius space,

$$\|\mathcal{L}_C - \mathcal{L}_C^\sharp\| \leq 2\kappa_F^2 \|C_-\|_1, \quad \kappa_F = \max_a \|F_a\|_2, \quad (6)$$

so the distance of $\mathcal{L}_C$ from the CPTP cone of generators is controlled by the negative eigenvalue mass $\|C_-\|_1 = \sum_{r:\lambda_r < 0} |\lambda_r|$ of its Kossakowski matrix.

3. The architectural parameterization $C = BB^\dagger$ of Sec. II C has $C_- = 0$ identically, so its CPTP violation is exactly zero for every value of the network weights B .

Part (3) is the statistical content of the “make invalid states unrepresentable” principle at the level of the *generator*: whereas a soft penalty drives $\|C_-\|_1$ toward—but not to—zero and thus incurs the residual violation (6), the Gram parameterization removes it by construction, mirroring at the generator level what the Cholesky map does at the state level (Theorem 1).

B. Sharpened residual-to-error stability

Theorem 4 bounds the trajectory error by the accumulated residual with constant one. That constant is optimal for a general CPTP semigroup, but real dissipative dynamics contract, and the sharper constant is the logarithmic norm of the generator in trace norm. We first record the relevant dissipativity, then the sharpened estimate.

Lemma 10 (Trace-norm logarithmic norm of a GKSL generator). *Let $\mathcal{L}$ be a GKSL generator and $\Phi_t = e^{t\mathcal{L}}$ its CPTP semigroup. Define the trace-norm logarithmic norm on the space $\mathcal{V}_0$ of Hermitian traceless operators,*

$$\mu(\mathcal{L}) = \lim_{h \downarrow 0} \frac{\|\text{id} + h\mathcal{L}\|_{\mathcal{V}_0} - 1}{h}, \quad (7)$$

where $\|\cdot\|_{\mathcal{V}_0}$ is the operator norm induced by $\|\cdot\|_1$ on $\mathcal{V}_0$. Then $\mu(\mathcal{L}) \leq 0$, and for every $X \in \mathcal{V}_0$ and $t \geq 0$, $\|\Phi_t X\|_1 \leq e^{\mu(\mathcal{L})t} \|X\|_1$. If in addition Φ_t has a unique full-rank stationary state ρ_∞ and is primitive (its spectral gap in the Hilbert–Schmidt sense is $g > 0$), then $\mu(\mathcal{L}) < 0$; more precisely there is a time $t_\star < \infty$ and a constant $\vartheta \in (0, 1)$ with $\|\Phi_{t_\star} X\|_1 \leq \vartheta \|X\|_1$ for all $X \in \mathcal{V}_0$, equivalently $\mu(\mathcal{L}) \leq t_\star^{-1} \ln \vartheta < 0$.

Theorem 11 (Dissipativity-sharpened a posteriori error bound). *Under the hypotheses of Theorem 4, with $\mu = \mu(\mathcal{L}) \leq 0$ the trace-norm logarithmic norm of Lemma 10, the error $e(t) = \hat{\rho}(t) - \rho(t)$ obeys, for all $t \geq 0$,*

$$\frac{1}{2} \|e(t)\|_1 \leq e^{\mu t} \frac{1}{2} \|e(0)\|_1 + \frac{1}{2} \int_0^t e^{\mu(t-s)} \|r(s)\|_1 \, ds. \quad (8)$$

Consequently, if $\mu < 0$ and $\sup_s \|r(s)\|_1 \leq \bar{r}$, then $\sup_{t \geq 0} \frac{1}{2} \|e(t)\|_1 \leq \frac{1}{2} \|e(0)\|_1 + \bar{r}/(2|\mu|)$, a bound uniform

in time: minimizing the physics residual controls the true error over the whole trajectory, not merely up to a horizon.

Equation (8) recovers Theorem 4 at the conservative value $\mu = 0$ and improves on it whenever the learned dynamics are strictly dissipative, which is the generic case for the open systems considered here; the constant-factor tightness of the $\mu = 0$ certificate is already visible in Table VII.

C. Approximation, generalization, and collocation complexity

We now decompose the total surrogate error into what the hypothesis class can represent and what finitely many collocation points can certify. Fix the level- N truncated generator $\tilde{\mathcal{L}}^{(N)}$ of Theorem 8, whose tail is η_N , and let the surrogate be trained by minimizing the empirical physics residual over a collocation sample.

Definition 1 (Population and empirical residual risk). *For a differentiable Hermitian trace-one path $\hat{\rho}$ on $[0, T]$ and a generator $\mathcal{G}$, define the population residual risk and its empirical counterpart on collocation nodes $\tau_1, \dots, \tau_{N_c}$ drawn i.i.d. uniformly on $[0, T]$,*

$$\mathcal{R}(\hat{\rho}; \mathcal{G}) = \frac{1}{T} \int_0^T \left\| \dot{\hat{\rho}}(\tau) - \mathcal{G}[\hat{\rho}(\tau)] \right\|_F^2 d\tau, \quad \hat{\mathcal{R}}(\hat{\rho}; \mathcal{G}) = \frac{1}{N_c} \sum_{\ell=1}^{N_c} \left\| \dot{\hat{\rho}}\left(\sup_{t \in [0, T]} \frac{1}{2} \|\hat{\rho}(t)\|_F\right) \right\|_F^2 \rho(t)_1 \leq \underbrace{\frac{1}{2} \Xi(\mu, T) C_\rho \eta_N}_{\text{truncation}} + \underbrace{\frac{\sqrt{T}}{2} \Xi(\mu, T) (\epsilon_{\text{approx}} + \epsilon_\rho)}_{\text{surrogate}} \quad (9)$$

Assumption 1 (Bounded, Lipschitz surrogate class). *The hypothesis class $\mathcal{F}$ of Cholesky-parameterized paths is such that every $\hat{\rho} \in \mathcal{F}$ and its time derivative are uniformly bounded, $\sup_\tau \|\hat{\rho}(\tau)\|_F \leq 1$ and $\sup_\tau \|\dot{\hat{\rho}}(\tau)\|_F \leq G$, the pointwise residual is bounded by $R_{\max} = (G + \Lambda)^2$ with Λ the operator norm of $\mathcal{G}$ on Hermitian operators, and the map $\tau \mapsto \hat{\rho}(\tau)$ is Lip-Lipschitz with a finite covering number $\log \mathcal{N}(\mathcal{F}, \varsigma) \leq \mathfrak{C} \log(1/\varsigma)$ for resolution $\varsigma > 0$ (finite-dimensional weights with bounded norm).*

Theorem 12 (Approximation–generalization decomposition and collocation complexity). *Let $\rho^{(N)}$ solve the level- N truncated master equation with generator $\mathcal{G} = \tilde{\mathcal{L}}^{(N)}$ and let $\mu \leq 0$ be its trace-norm logarithmic norm (Lemma 10). Suppose the surrogate $\hat{\rho} = \rho_\phi$ is trained under Assumption 1 to empirical residual risk $\hat{\mathcal{R}}(\rho_\phi; \mathcal{G}) \leq \hat{\mathcal{R}}_\star$ with matched initial condition $\rho_\phi(0) = \rho^{(N)}(0)$. Then with probability at least $1 - \delta$ over the i.i.d. collocation draw, the true trajectory error of the surrogate for the truncated dynamics obeys*

$$\sup_{t \in [0, T]} \frac{1}{2} \left\| \rho_\phi(t) - \rho^{(N)}(t) \right\|_1 \leq \frac{\sqrt{T}}{2} \left(\underbrace{\hat{\mathcal{R}}_\star^{1/2}}_{\text{empirical fit}} + \underbrace{\epsilon_{\text{gen}}(N_c, \delta)}_{\text{generalization}} \right) \quad (10)$$

where $\Xi(\mu, T) = \int_0^T e^{\mu s} ds \leq \min\{T, 1/|\mu|\}$ and the generalization gap satisfies

$$\epsilon_{\text{gen}}(N_c, \delta)^2 \leq \frac{c R_{\max}}{\sqrt{N_c}} \left(\sqrt{\mathfrak{C} \log N_c} + \sqrt{\log(1/\delta)} \right) \quad (11)$$

for a universal constant c . In particular, to guarantee $\hat{\mathcal{R}}_\star^{1/2} + \epsilon_{\text{gen}} \leq \epsilon$ with the approximation floor $\hat{\mathcal{R}}_\star^{1/2} = \epsilon_{\text{approx}}$ achievable by Theorem 2, it suffices to take a collocation budget

$$N_c = \tilde{O} \left(\frac{R_{\max}^2 (\mathfrak{C} + \log(1/\delta))}{(\epsilon - \epsilon_{\text{approx}})^4} \right). \quad (12)$$

The decomposition (10) is the central surrogate statement: the true (out-of-sample) trajectory error is bounded by a term the network can drive down by expressivity (Theorem 2) plus a term that is paid off by collocation points at the rate (11), and the dissipative factor $\Xi(\mu, T)$ caps the accumulation exactly as in Theorem 11. It converts “small training residual” into “small certified error at unseen times,” which is what an emulator must promise.

Corollary 13 (Truncation-plus-surrogate error and the cost-optimal band). *Under the hypotheses of Theorems 8 and 12, the surrogate’s error against the exact (full-band) trajectory satisfies, with probability at least $1 - \delta$,*

with $C_\rho = \sup_s \|\tilde{\rho}(s)\|_1$. If the tail decays geometrically, $\eta_N \leq \eta_0 \beta^N$ with $\beta \in (0, 1)$, and the per-level compute cost grows as $\text{cost}(N) = c_0(2N+1)$, then the band that minimizes the certified error at fixed budget, or the cost at fixed error tolerance ϵ_{tol} above the surrogate floor, is

$$N_\star = \left\lceil \frac{\ln(\eta_0 \Xi(\mu, T) C_\rho / \epsilon_{\text{tol}})}{\ln(1/\beta)} \right\rceil, \quad (14)$$

beyond which additional Fourier modes reduce the certificate below the surrogate floor and cannot improve the guaranteed accuracy.

D. Predictive uncertainty and calibrated coverage

Finally we quantify the surrogate’s *uncertainty*. Treating identification as Bayesian inference over the generator parameters Θ turns the point estimate of Theorem 5 into a predictive distribution whose credible bands we show are asymptotically calibrated.

Assumption 2 (Regular Gaussian observation model). *Observations are $y_{ij} = \text{Tr}[O_j \rho_{\Theta_\star}(t_i)] + \sigma \epsilon_{ij}$ with i.i.d. standard-Gaussian ϵ_{ij} and known $\sigma > 0$, over $M = np$ measurements (n time-observable configurations replicated so $M/n \rightarrow \infty$). The map $\Theta \mapsto f(\Theta) =$*

$(\text{Tr}[O_j \rho_\Theta(t_i)])_{ij}$ is C^2 on a neighborhood of the interior true value Θ_* , and the sensitivity Gramian $\mathcal{I}(\Theta_*) = \sigma^{-2} J(\Theta_*)^\top J(\Theta_*)$ is nonsingular, with J as in Theorem 5. The prior π has a continuous density that is positive at Θ_* .

Theorem 14 (Posterior contraction and calibrated predictive coverage). *Under Assumption 2, the posterior $\Pi(\cdot | y)$ over Θ satisfies:*

1. Contraction. *For every $M_n \rightarrow \infty$, $\Pi(\|\Theta - \Theta_*\|_2 > M_n n^{-1/2} | y) \rightarrow 0$ in probability; the posterior concentrates on ℓ_2 -balls of radius $O(n^{-1/2})$ around the truth.*
2. Bernstein–von Mises. *The total-variation distance between the posterior of $\sqrt{n}(\Theta - \hat{\Theta}_n)$ and the Gaussian $\mathcal{N}(0, [n^{-1}\mathcal{I}(\Theta_*)]^{-1})$ tends to zero in probability, where $\hat{\Theta}_n$ is the maximum-likelihood estimate.*
3. Calibrated coverage. *Consequently any level- $(1 - \alpha)$ posterior credible set $\mathcal{C}_{1-\alpha}$ for Θ has frequentist coverage $\Pr(\Theta_* \in \mathcal{C}_{1-\alpha}) \rightarrow 1 - \alpha$; and for any fixed (t, O) the induced predictive interval for the observable $\text{Tr}[O \rho_\Theta(t)]$, obtained by push-forward of $\Pi(\cdot | y)$, has asymptotic coverage $1 - \alpha$. The interval half-width scales as $z_{1-\alpha/2} \sigma \|J_{tO} [J^\top J]^{-1/2}\|_2$, shrinking at rate $n^{-1/2}$ and inflating exactly along the sloppy directions where the smallest singular value $s_{\min}$ of J is small (Theorem 5).*

Theorem 14 makes the surrogate a genuinely probabilistic emulator: it emits not just a trajectory but calibrated error bars, and part (3) ties the width of those bars back to the identifiability Gramian, so the same singular direction that makes a rate “sloppy” (the qutrit dephasing γ_ϕ of Sec. VD) is automatically reported as a wide predictive interval rather than a silently overconfident one. This is the uncertainty-quantification counterpart of the deterministic identifiability bound of Theorem 5 and closes the surrogate-theory loop: expressivity (Theorem 2), physical validity (Theorems 1, 3, Proposition 9), residual-controlled accuracy (Theorems 4, 11), collocation complexity (Theorem 12), truncation trade-off (Corollary 13), and calibrated uncertainty (Theorem 14).

V. SIMULATIONS AND RESULTS

All numerical values below are produced by the `ctpcpinn` package in the repository’s `code/` directory and written directly to the figure and table files; none are entered by hand. Simulations run on CPU using PyTorch and SciPy [16, 17], with Adam [34] for the physics-informed runs. Each training experiment is repeated over five independent seeds—each seed redrawing the network initialization and the measurement-noise realization—and reported as the mean $\pm$ a 95% Student- t confidence interval (CI) over seeds; the compiler and

spectral-truncation benchmarks are deterministic and reported as the median over repeated timings. We compare, as the experiment dictates, the hard-constrained CPTP-PINN and its spectral-frame variant (this work); plain-MLP and Fourier-feature density networks; a soft-penalty PINN (an eigenvalue-negativity penalty, the soft alternative argued against in Sec. VA); a fully unconstrained network; and a classical least-squares fit using the exact model structure. The aim is to test the structural claims—exact physicality, spectral-bias-free reconstruction, parameter and trajectory recovery, robustness under sparse observations, generalization of the learned generator, operator-system truncation convergence, and compiler cost—on small, fully reproducible problems, not to claim production-scale or hardware performance.

A. Hard constraints versus soft penalties

The Bloch-sphere picture shows why a penalty is not a guarantee: a qubit matrix $\rho(r) = \frac{1}{2}(I + r_x \sigma_x + r_y \sigma_y + r_z \sigma_z)$ is a density matrix iff $\|r\|_2 \leq 1$, and an unconstrained $r(t)$ can leave the ball even when the loss eventually pushes it back. During optimization such excursions corrupt residuals, produce negative probabilities, and interact badly with parameter estimation. We test this directly by including a soft-penalty PINN [a positivity penalty $\sum_i \text{ReLU}(-\lambda_i(\rho))$ added to the loss] alongside the unconstrained baseline. The positivity-violation column of Table I and the sparse-data results of Table II quantify the gap: the Cholesky map never violates positivity, whereas the soft and unconstrained surrogates do, and the difference becomes decisive when observations are scarce.

B. Single-qubit system identification

We consider a driven qubit with amplitude damping and pure dephasing,

$$H = \frac{\omega}{2} \sigma_z + \frac{\Omega}{2} \sigma_x, \quad (15)$$

$$\begin{aligned} \mathcal{L}[\rho] = & -i[H, \rho] + \gamma_1 (\sigma_- \rho \sigma_+ - \frac{1}{2} \{\sigma_+ \sigma_-, \rho\}) \\ & + \frac{\gamma_\phi}{2} (\sigma_z \rho \sigma_z - \rho), \end{aligned} \quad (16)$$

with ground truth $\omega = 2\pi(0.5)$, $\Omega = 2\pi(0.1)$, $\gamma_1 = 0.30$, $\gamma_\phi = 0.15$, initial state $\rho_0 = |+\rangle\langle +|$, and noisy samples of $\langle \sigma_x \rangle, \langle \sigma_y \rangle, \langle \sigma_z \rangle$ on a uniform grid over $t \in [0, 3]$ with additive Gaussian noise of standard deviation 0.02. All four methods jointly fit the four generator parameters and the trajectory.

Table I reports the relative parameter-recovery errors, mean state fidelity, and positivity-violation rate, each as mean $\pm$ 95% CI over five seeds. Three points stand out. First, the hard constraint is decisive *among the neural methods*: the CPTP-PINN recovers every parameter far more accurately than the soft-penalty and unconstrained

networks (for example the dephasing rate γ_ϕ to 13% versus more than 140%), reaches mean state fidelity 1.0000, and—by construction—never leaves the physical set (0 positivity violations, versus 0.4–0.6% of time points for the soft and unconstrained surrogates, which the eigenvalue penalty does not prevent). Second, a classical least-squares fit that already knows the exact model structure remains the strongest method on this dense-data task, recovering the dissipative rates to $\approx 2\%$; where a low-parameter model and full data are available, model-based fitting is hard to beat, and the CPTP-PINN matches it on the Hamiltonian parameters (ω to 0.1%, Ω to 2.6%) and on fidelity without surpassing it. The value of the architecture is therefore guaranteed physicality and robustness, not beating a well-specified classical fit on dense data; the next experiment isolates the regime where that matters. Figure 2 shows the parameter-recovery errors and the per-time state fidelity with confidence intervals.

C. Robustness under sparse measurements

Using the same qubit system, we randomly retain only a fraction of the observation entries and measure how trajectory recovery degrades, comparing the CPTP-PINN, the soft-penalty PINN, and the unconstrained network at observation fractions 1.0, 0.5, 0.25, and 0.10. As shown in Table II and Fig. 3(a), the constrained model keeps the mean trace distance to the exact trajectory between 0.004 and 0.010 across all fractions—one to two orders of magnitude below the soft and unconstrained baselines (both ≈ 0.37 , with confidence intervals as wide as their means, reflecting frequent failures)—and degrades only mildly as observations are removed down to 10%. This is the regime the architectural constraint is designed for: with few observations the unconstrained surrogate can drift outside the positive cone, the soft penalty competes with the data and residual terms, and only the Cholesky map is physical by construction at every point.

D. Spectral-frame reconstruction of fast qutrit leakage dynamics

We next stress-test the method on a transmon-like three-level system with anharmonic Hamiltonian $H = E_1 |1\rangle\langle 1| + E_2 |2\rangle\langle 2|$ at $E_1 = 2\pi(2.0)$, $E_2 = 2\pi(3.8)$, and cascade decay plus dephasing $L_{10} = \sqrt{\gamma_{10}} |0\rangle\langle 1|$, $L_{21} = \sqrt{\gamma_{21}} |1\rangle\langle 2|$, $L_\phi = \sqrt{\gamma_\phi} \text{diag}(0, 1, 2)$, with true rates $\gamma_{10} = 0.5$, $\gamma_{21} = 0.2$, $\gamma_\phi = 0.1$. Populations and coherence quadratures are observed over $t \in [0, 4]$ with 2% noise. The Bohr frequencies of H ($\omega_{10} \approx 12.6$, $\omega_{21} \approx 11.3$, $\omega_{20} \approx 23.9$) drive fast coherent oscillations on top of slow dissipative decay—about fifteen full oscillations over the window—and this is the regime that defeats a smooth-in-time density network. *Direct* reconstruction with a plain MLP ($K = 0$) or a deterministic Fourier-feature encoding ($K = 48$) was the principal open problem of earlier

versions of this work: across five seeds it is unreliable (Table III, Fig. 3(b)), with mean state fidelity 0.77 ± 0.35 for the plain MLP and 0.52 ± 0.49 for the Fourier network—intervals so wide that neither encoding reconstructs the fast dynamics robustly.

The spectral-frame (interaction-picture) density network removes the obstacle at the architecture level: it learns only the slow envelope $\tilde{\rho}_\phi(t) = U(t)^\dagger \rho_\phi(t) U(t)$ (here $\sim 40\times$ slower than the lab trajectory) through the same Cholesky map, and supplies the fast Bohr oscillation analytically through the known $U(t) = e^{-iHt}$. With *no* Fourier features and an otherwise identical network, it reconstructs the trajectory reliably: mean state fidelity 0.985 ± 0.002 across the five seeds, more than two orders of magnitude tighter in its confidence interval than either baseline, while remaining a valid density matrix by construction (Theorem 7). Reliable reconstruction in turn makes the inverse problem well posed: an integral least-squares readout of the linear-in-rates generator from the learned envelope (Appendix D) recovers the dominant decay rates γ_{10}, γ_{21} to mean relative error 0.273 ± 0.025 , whereas the same readout applied to the unreconstructed plain and Fourier trajectories is uninformative (those trajectories never enter the regime where the readout is meaningful). The dephasing rate γ_ϕ remains a sloppy direction under 2% measurement noise—consistent with the local identifiability bound (Theorem 5)—but on the *noiseless* envelope the readout is exact, isolating the residual gap to measurement noise rather than spectral bias. Fast multilevel reconstruction, previously our principal open problem, is thus resolved by the spectral frame, and rate identification is reduced to a measurement-noise question.

E. A transferable two-qubit dissipative gate

The two-qubit experiment uses a time-dependent exchange coupling $J(t) = J_0 \exp[-(t - 1.5)^2 / (2 \cdot 0.3^2)]$ with $J_0 = 2\pi(0.05)$, local detunings $\Delta_1 = 2\pi(5.0)$, $\Delta_2 = 2\pi(4.8)$, and local amplitude damping and dephasing on each qubit; Fourier time-features handle the fast detuning dynamics. The model is trained on the $|00\rangle$ initial state. The scientifically meaningful test is whether the *learned generator*—rather than the single trained trajectory—predicts held-out initial states, so we propagate the recovered Lindbladian from $|01\rangle$, $|10\rangle$, and $|+0\rangle$ and compare to the exact dynamics. The network reproduces the trained $|00\rangle$ trajectory at fidelity 0.994 ± 0.001 , and the recovered generator generalizes to the three held-out states at mean fidelities 0.95, 0.96, and 0.97 (overall average 0.97 ± 0.005 ; Table IV, Fig. 3(c)). In contrast to the direct qutrit reconstruction of Sec. VD, this result is stable across seeds (note the narrow confidence intervals), because the test propagates the learned *generator* with an exact solver rather than reading off the neural trajectory. This is the property that matters for device characterization—a transferable open-system model

TABLE I. **Single-qubit system identification (Experiment 1).** Parameter-recovery relative error, mean state fidelity, and positivity-violation rate (fraction of time points with a negative eigenvalue) for the four methods. Values are mean $\pm$ 95% confidence interval (CI) over 5 seeds. True values: $\omega = 3.1416$, $\Omega = 0.6283$, $\gamma_1 = 0.3000$, $\gamma_\phi = 0.1500$. The classical least-squares fit uses the exact model structure and is the strongest method on this full-data task; the PINN advantage appears under sparse measurements (Table II).

Method	ω err	Ω err	γ_1 err	γ_ϕ err	Mean fidelity	Pos. viol.
CPTP-PINN (ours)	0.001 ± 0.001	0.026 ± 0.013	0.079 ± 0.027	0.131 ± 0.050	1.0000 ± 0.0000	0.000
Soft-penalty PINN	0.158 ± 0.195	0.358 ± 0.287	0.266 ± 0.288	1.455 ± 1.184	0.9507 ± 0.1163	0.004
Unconstrained	0.157 ± 0.195	0.362 ± 0.284	0.276 ± 0.283	1.439 ± 1.185	0.9511 ± 0.1165	0.006
Classical LSQ	0.001 ± 0.001	0.016 ± 0.012	0.019 ± 0.008	0.020 ± 0.019	1.0000 ± 0.0000	0.000

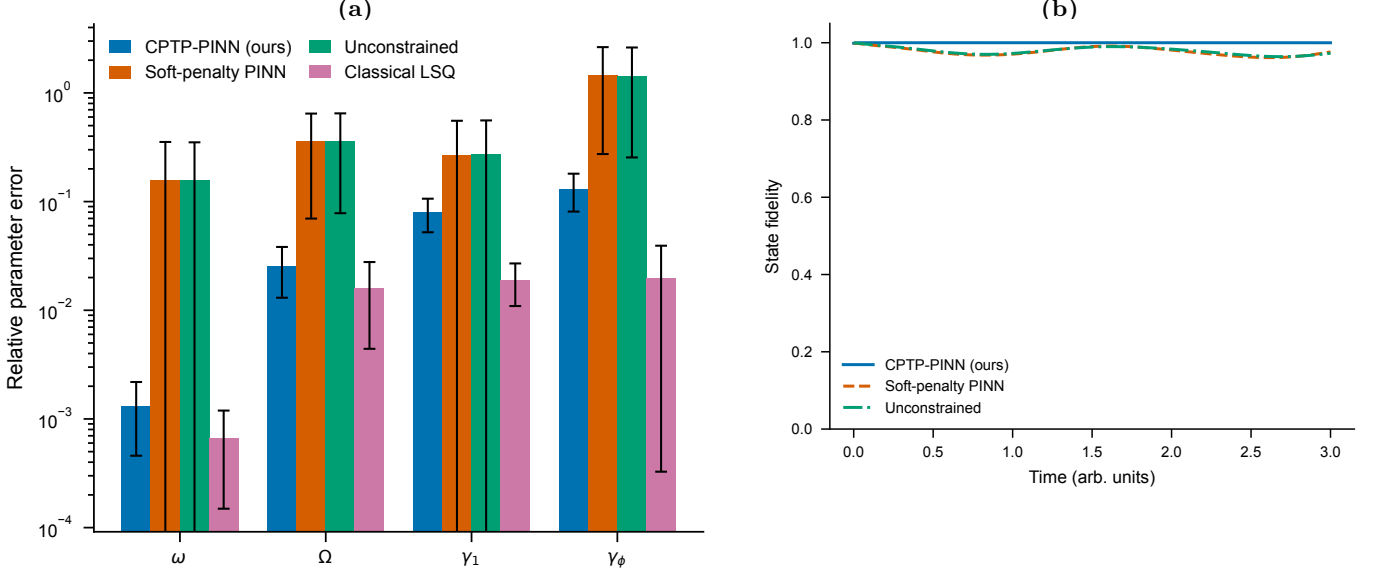


FIG. 2. **Single-qubit system identification.** (a) Relative parameter-recovery error per method (mean and 95% confidence interval over five seeds, log scale): the CPTP-PINN strongly outperforms the soft-penalty and unconstrained networks and is competitive with a classical least-squares fit that already knows the exact model (Table I). (b) State fidelity between the learned and exact density-matrix trajectories (representative seed); the hard-constrained CPTP-PINN stays inside $\mathcal{D}(\mathcal{H})$ by construction. Error bars and bands are 95% confidence intervals over five random seeds.

TABLE II. **Robustness under sparse measurements (Experiment 2).** Mean trace distance to the exact single-qubit trajectory versus retained observation fraction, for the three neural methods. Mean $\pm$ 95% CI over 5 seeds.

Fraction	CPTP-PINN (ours)	Soft-penalty	Unconstrained
1.00	0.0044 ± 0.0038	0.3668 ± 0.3497	0.3688 ± 0.3482
0.50	0.0052 ± 0.0037	0.3683 ± 0.3469	0.3717 ± 0.3436
0.25	0.0050 ± 0.0018	0.3682 ± 0.3490	0.3709 ± 0.3467
0.10	0.0100 ± 0.0062	0.3572 ± 0.3616	0.3572 ± 0.3626

rather than a memorized curve—and it holds because physicality and CPTP structure are enforced by construction while the few generator parameters are shared across all initial states.

F. Matrix-free compiler scaling

The compiler experiment compares dense superoperator evaluation against structured matrix-free residual evaluation on random Lindblad systems: dense mode forms the $d^2 \times d^2$ Liouvillian, structured mode evaluates commutator and dissipator terms directly from H and the jump operators. The robust, hardware-independent conclusion is the *memory* separation (Table V, Fig. 4): consistent with Theorem 6, dense storage scales as d^4 , whereas the structured representation stores only H , the m jump operators, and ρ , i.e. $O(md^2)$, so its memory advantage widens with d . The two residuals agree to floating-point precision, so structured evaluation is exact, not an approximation. We do *not* claim a speed advantage: our benchmark’s dense path reconstructs the full Liouvillian on every state evaluation (an $O(md^4)$ build paid per state), so the wall-clock gap it reports reflects that implementation artifact rather than an intrinsic per-

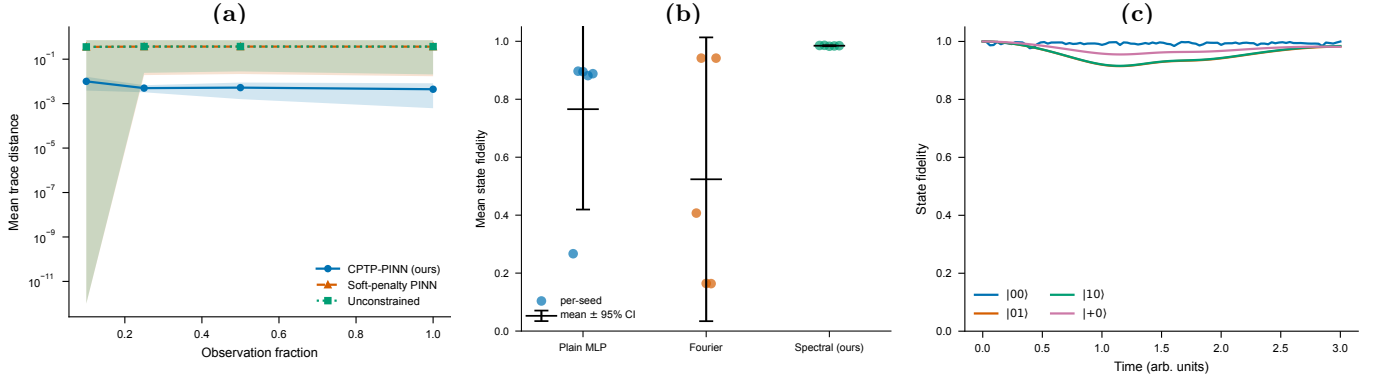


FIG. 3. **Robustness, the spectral-frame cure, and generalization across systems.** (a) Sparse-measurement ablation on the single qubit: mean trace distance to the exact trajectory versus retained observation fraction (log scale), for the CPTP-PINN, a soft-penalty PINN, and an unconstrained network. (b) Fast qutrit reconstruction: per-seed mean state fidelity for the plain MLP, the Fourier-feature density network, and the spectral-frame (interaction-picture) density network. The plain and Fourier baselines are unreliable across seeds, whereas the spectral frame reconstructs the fast dynamics reliably (tight interval near unity)—resolving the principal open problem. (c) Two-qubit dissipative gate: state fidelity for the trained $|00\rangle$ trajectory and for three held-out initial states obtained by propagating the learned generator (representative seed). Error bars/bands and points are over five random seeds; see Table II and Tables III and IV.

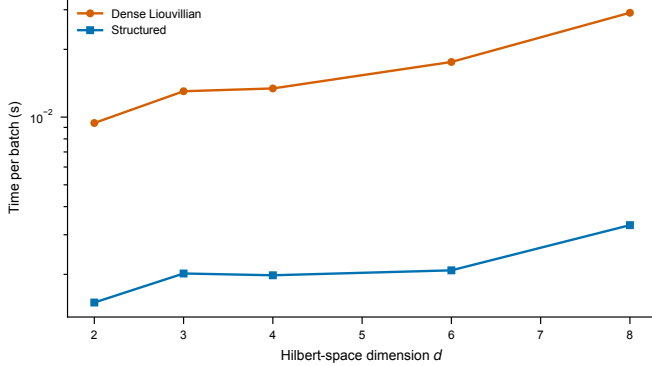


FIG. 4. **Matrix-free compiler scaling.** Dense superoperator residual-evaluation time versus structured matrix-free evaluation (median over 50 repeats, log scale). The structured path avoids materializing the $d^2 \times d^2$ Liouvillian; its robust advantage is memory [$O(md^2)$ versus $O(d^4)$], not speed—the dense timings shown rebuild the Liouvillian per state and so overstate any wall-clock gap (see text and Table V).

matvec speedup, and amortizing the Liouvillian build across states would largely remove it. We therefore report timing only as an implementation detail and treat the $O(d^4)$ -versus- $O(md^2)$ memory scaling as the genuine, asymptotic separation.

G. Operator-system spectral truncation of the generator

The final experiment realizes the operator-system spectral truncation of Theorem 8 as a numerical object. A driven, dissipative two-qubit system—large local detunings $\Delta_1 = 2\pi(5.0)$, $\Delta_2 = 2\pi(4.8)$, a strong time-

dependent exchange coupling $J(t)(XX+YY)$ with $J_0 = 2\pi(0.4)$, and local amplitude damping—is moved into the interaction picture set by the detunings, where the generator $\hat{\mathcal{L}}_t$ is almost periodic. We extract its Fourier modes $\{\mathcal{L}_k\}$, keep the band $|k| \leq N$, propagate the level- N truncated generator (re-rotating the envelope to the lab frame), and measure the mean state fidelity to the exact trajectory together with the out-of-band weight $\eta_N = \sum_{|k|>N} \|\mathcal{L}_k\|$ that certifies the generator-approximation error. Figure 5 and Table VI show a controlled multi-resolution hierarchy: the secular (Davies) generator $N=0$ already captures the slow envelope at mean fidelity 0.76; the rotating-wave approximation $N=1$ reaches 0.96; and the fidelity converges monotonically to floating-point precision as the band is widened (0.99999 by $N=4$), with η_N decreasing in lock-step. The full band reproduces the structured kernel exactly, so the truncation interpolates—with an a posteriori certificate at every level—between the cheapest time-independent secular model and the exact generator. This is the spectral-truncation construction of noncommutative geometry and C^* -algebraic kernel machines [40, 42], here instantiated for a completely positive dynamical generator.

VI. DISCUSSION AND CONCLUSION

The experiments separate what the architectural ingredients—the hard density constraint, the spectral frame, and the matrix-free compiler—each contribute, and where the approach still falls short. We discuss these in turn, then the optimization geometry that governs all three.

What the spectral frame buys. The hard density constraint guarantees a *physical* trajectory but not an *accu-*

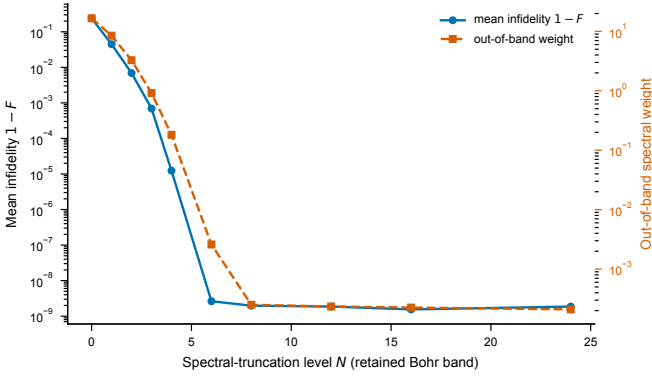


FIG. 5. **Operator-system spectral truncation of the open-system generator.** Mean infidelity $1 - F$ of the propagated level- N truncated interaction-picture generator (left axis) and the out-of-band spectral weight $\eta_N = \sum_{|k|>N} \|\mathcal{L}_k\|$ of Theorem 8 (right axis, log scale) versus the retained Bohr-band level N , for a driven dissipative two-qubit system. $N=0$ is the secular (Davies) generator, $N=1$ the rotating-wave approximation, and the full band recovers the exact generator (and the structured kernel) to floating-point precision. The certificate η_N tracks and bounds the infidelity, decreasing monotonically to zero. Deterministic (no seed averaging).

rate one: on fast multilevel dynamics a smooth density network is biased toward low frequencies and cannot resolve the Bohr oscillations, which was the principal failure of the earlier approach. The spectral frame separates the two scales: the unitary $U(t) = e^{-iHt}$ carries the fast, exactly-known oscillation analytically, and the network learns only the slow envelope through the same hard map. The change is not a new loss or a tuned encoding but a change of *representation*—the eigen-operator decomposition of the dynamics—so it inherits the physicality guarantee (Theorem 7) while removing spectral bias. Empirically this turns an unreliable reconstruction (mean fidelity ≈ 0.5 – 0.8 , intervals as wide as the means) into a reliable one (0.985 ± 0.002) and, by conditioning the envelope derivative, reduces rate identification to a measurement-noise problem rather than an optimization one. The same eigen-operator decomposition supplies the operator-system spectral truncation: band-limiting the generator to its slow Bohr modes is, mathematically, the spectral truncation by which noncommutative geometry approximates a C^* -algebra by finite operator systems [40, 42], now applied to a completely positive dynamical generator and equipped with a computable error certificate. The secular (Davies) and rotating-wave approximations emerge as its coarsest two levels, and the exact generator as the full band—so a single dial trades cost for fidelity with an a posteriori guarantee at every setting.

What the hard constraint buys. The Cholesky map changes the feasible set. A soft penalty for positivity competes with data fit, residual fit, and optimizer noise; a hard parameterization removes that competition. On the full-data single-qubit problem this is already visible

in the *neural* comparison: the CPTP-PINN recovers every parameter far more accurately than the soft-penalty and unconstrained networks and never leaves the physical set, while those baselines do. The only method that beats it there is a classical least-squares fit that already knows the exact model, whose advantage is confined to the dense-data, low-parameter setting. The architectural constraint pays off most clearly in two places the experiments isolate: it guarantees a physical trajectory at every step (zero positivity violations), and it degrades gracefully under sparse measurements (Sec. V C), where the soft and unconstrained surrogates leave or fight the positive cone. In larger systems, where the positive cone is a small subset of Hermitian trace-one matrices, this gap is expected to widen.

What the compiler buys. The compiler determines whether the residual can be evaluated without forming an object whose size scales as d^4 . In a multi-qubit Hilbert space $d = 2^n$, so dense Liouvillian storage scales as 16^n ; structured kernels still face exponential Hilbert-space growth but avoid an unnecessary square of the density-vector dimension. This is the difference between a prototype that only runs on toy models and a path that can exploit locality, batching, sparsity, tensor products, and GPU memory hierarchy. The most direct accelerator path lowers each Hamiltonian and dissipator term to batched matrix multiplications, fuses anticommutator terms, and keeps density matrices resident on the GPU; circuit- or pulse-level noise hypotheses can be compiled into Hamiltonian and jump-operator terms via existing stacks [18, 19], and resource-estimation tools may interact with learned models [35, 36].

Optimization geometry and failure modes. The Cholesky map is nonlinear and redundant; near rank-deficient states, small changes in the factor can produce small eigenvalues and ill-conditioning. This is a reason for careful initialization, diagonal floors, learning-rate schedules, and low-rank-plus-isotropic variants, not a reason to abandon the representation. The framework has three distinct diagnostics that should not be conflated (Appendix L): state physicality (guaranteed), residual size, and held-out prediction error. A model can be physical at every time and still solve the wrong dynamics; only the residual and held-out validation certify the model, and only an identifiability analysis (Theorem 5) certifies the parameters.

A. Limitations

1. *Rate sloppiness under measurement noise.* The spectral frame resolves the fast-reconstruction problem (Sec. V D), so what remains is a measurement-noise question: with the trajectory reliably reconstructed, the dominant decay rates are recovered, but a weakly excited rate (the qutrit dephasing γ_ϕ) is a sloppy direction at 2% noise, exactly as the local identifiability bound (Theorem 5) predicts—the readout is exact on

the noiseless envelope. Richer measurement designs (active experiment design that maximizes the smallest singular value of the sensitivity Gramian) and derivative denoising are the natural next steps.

2. *Known frame Hamiltonian.* The spectral frame and the operator-system truncation use the eigenbasis of a *known* coherent part H . When H itself is unknown or strongly time-dependent, the frame must be co-learned or chosen adaptively; a partial frame (the dominant static part) still removes the fast oscillations, as in the two-qubit experiments.
3. *No hardware data.* The experiments are synthetic CPU simulations; they test the implementation and theory but do not demonstrate performance on a quantum device.
4. *Markovian assumption.* The framework assumes GKSL dynamics. Non-Markovian systems require extended state spaces, memory kernels, or process tensors.
5. *Identifiability is measurement-dependent.* The theorems cannot overcome uninformative observables: if the sensitivity Gramian is rank deficient, no optimizer can recover all parameters.
6. *Scope of the empirical claims.* The study is a small proof of concept (five seeds per experiment); it is not benchmarked against every modern system-identification method, and it does not claim state of the art.

Outlook: identification as calibrated statistical inference. Viewed through the lens that a quantum device is a stochastic object accessed only by sampling, open-system identification is a statistical-inference problem, and the constructions here are the parts of that pipeline where domain structure does the statistical work. The surrogate is a probabilistic emulator of an expensive simulator (or, ultimately, of hardware): the physics residual is a mechanistic prior that integrates the GKSL equation of motion directly into the estimator, so learning proceeds from limited, noisy data in the interpretable, knowledge-constrained regime rather than from an unconstrained black box. The hard Cholesky map is an exact structural constraint that removes an entire family of failure modes from the hypothesis class before any data are seen, and the operator-system spectral truncation supplies an a posteriori certificate that plays the role a sampled confidence interval would, but deterministically. The complementary statistical theory (Sec. IV) makes the estimation reading explicit: a collocation-sample complexity, a posterior that contracts at the parametric $n^{-1/2}$ rate, and asymptotically calibrated credible bands under the identifiability Gramian of Theorem 5. That same Gramian turns the residual sloppiness we observe (the weakly excited qutrit dephasing rate) into a concrete design objective: choosing measurements that maximize its smallest singular value is active, uncertainty-aware experiment

design, the natural next step toward closing the loop between a learned open-system model, its calibrated uncertainty, and the next experiment to run. The broader program to which this paper belongs treats observable estimation, tomography, certification, decoding, and drift tracking as one statistical thread; the present contribution is its structure-preserving, surrogate-modeling entry.

Conclusion. Structure-preserving, spectrally framed PINNs identify open quantum dynamics by uniting a physical state parameterization, a spectral-frame eigen-operator representation, an operator-system spectral truncation of the generator, generator learning that preserves GKSL form, and matrix-free residual compilation. The conceptual point is twofold: physicality should be an architectural invariant rather than a penalty applied after the fact, and the fast coherent structure of open-system dynamics should be carried by an analytic change of frame rather than learned, so that the network represents only what is genuinely unknown. The mathematics shows that these choices do not destroy universal approximation, that the learned generator remains CPTP, that the spectral frame is exactly physical and exactly equivalent to the lab dynamics, that residuals and the spectral-truncation tail provide a posteriori certificates, and that structured compilation avoids dense Liouvillian storage. The spectral truncation moreover gives a controlled bridge—the secular and rotating-wave approximations as its coarsest levels—between the cheapest time-independent generator and the exact one. The experiments are small in scale and reported with five-seed statistics. Future work should extend to larger multi-qubit systems, co-learned or adaptive frames, hardware-calibration datasets, non-Markovian embeddings, and GPU-native kernels; the most promising engineering direction is a compiler that takes a high-level noise model and emits fused, batched, spectrally truncated, differentiable kernels for accelerator execution.

DATA AND CODE AVAILABILITY

This study uses no external datasets. All numerical values, tables, and figures are generated from fixed random seeds by the accompanying `ctpcpinn` package, which implements the method and regenerates every reported number; it will be released in a public repository upon publication. The reproduction recipe is given in Appendix N.

Appendix A: GKSL dynamics and the dense Liouvillian

Let $\rho(t) \in \mathcal{D}(\mathcal{H})$. A time-independent Markovian master equation has the form

$$\frac{d\rho}{dt} = \mathcal{L}_\Theta[\rho] = -i[H_\Theta, \rho] + \sum_{k=1}^m \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right), \quad (\text{A1})$$

with $H_\Theta = H_\Theta^\dagger$, $\gamma_k \geq 0$; the GKSL theorem characterizes generators of quantum dynamical semigroups in finite dimensions [1–4]. Using the column-major identity $\text{vec}(A\rho B) = (B^T \otimes A) \text{vec}(\rho)$, the dense Liouvillian is

$$\mathbb{L}_\Theta = -i(I \otimes H_\Theta - H_\Theta^T \otimes I) + \sum_k \gamma_k \left(L_k^* \otimes L_k - \frac{1}{2} I \otimes L_k^\dagger L_k - \frac{1}{2} (L_k^\dagger L_k)^T \otimes I \right), \quad (\text{A2})$$

which stores d^4 complex scalars, whereas direct evaluation of Eq. (A1) stores H and the L_k , i.e. $O(md^2)$.

Appendix B: Generator parameterization

For known jump operators with unknown rates,

$$\gamma_k(\xi_k) = \text{softplus}(\xi_k) + \gamma_{\min}, \quad \gamma_{\min} \geq 0; \quad (\text{B1})$$

for a full Kossakowski matrix in a fixed basis $\{F_a\}_{a=1}^q$, $C_\eta = B_\eta B_\eta^\dagger \succeq 0$ and

$$\mathcal{L}_\Theta[\rho] = -i[H_\Theta, \rho] + \sum_{a,b=1}^q (C_\eta)_{ab} \left(F_a \rho F_b^\dagger - \frac{1}{2} \{F_b^\dagger F_a, \rho\} \right), \quad (\text{B2})$$

with $H_\Theta(t) = \sum_\ell \theta_\ell(t) G_\ell$, $G_\ell = G_\ell^\dagger$.

Appendix C: Observation model and loss

With observables $\{O_j\}_{j=1}^p$, measurements y_{ij} at time t_i , and masks $m_{ij} \in \{0, 1\}$,

$$\mathcal{J}_{\text{data}}(\phi) = \frac{1}{\sum_{i,j} m_{ij}} \sum_{i,j} m_{ij} (\text{Tr}[O_j \rho_\phi(t_i)] - y_{ij})^2, \quad (\text{C1})$$

$$\mathcal{J}_0(\phi) = \|\rho_\phi(0) - \rho_0\|_F^2,$$

and $\mathcal{J}_{\text{phys}}$ as in Eq. (4). The derivative $\dot{\rho}_\phi$ is obtained by automatic differentiation; the centered finite-difference surrogate $\dot{\rho}_\phi(t) \approx [\rho_\phi(t+h) - \rho_\phi(t-h)]/2h$ is an alternative. The soft-penalty baseline adds $\lambda_{\text{pos}} \sum_i \text{ReLU}(-\lambda_i(\rho))$ to the loss. We use Adam [34]; L-BFGS is a natural second stage [37].

Appendix D: Spectral-frame parameterization and interaction-picture residual

Let $H = \sum_a E_a |a\rangle\langle a|$ be the (known) frame Hamiltonian in its eigenbasis and $U(t) = e^{-iHt}$. The spectral-frame density network outputs an envelope $\tilde{\rho}_\phi(t) \in \mathcal{D}(\mathcal{H})$ in the eigenbasis through the same Cholesky map [Eq. (2)], and the lab state is the analytic rotation

$$\rho_\phi(t) = U(t) \tilde{\rho}_\phi(t) U(t)^\dagger, \quad (\text{D1})$$

$$(\tilde{\rho}_\phi)_{ab} \mapsto e^{-i\omega_{ab}t} (\tilde{\rho}_\phi)_{ab} \quad (\text{Hadamard}),$$

with Bohr frequencies $\omega_{ab} = E_a - E_b$; for diagonal H the rotation is the entrywise phase $e^{-i\omega_{ab}t}$. Because $U(t)$ is unitary, $\rho_\phi(t) \in \mathcal{D}(\mathcal{H})$ by Theorem 7. The data and initial-condition terms [Eq. (C1)] use ρ_ϕ ; the physics residual is evaluated in the interaction picture,

$$\mathcal{J}_{\text{phys}}^{\text{spec}} = \frac{1}{N_c} \sum_\ell \left\| \dot{\tilde{\rho}}_\phi(\tau_\ell) - \tilde{\mathcal{L}}_{\Theta, \tau_\ell}[\tilde{\rho}_\phi(\tau_\ell)] \right\|_F^2, \quad (\text{D2})$$

$$\tilde{L}_k(t) = e^{i\omega_{ab}t} (L_k)_{ab},$$

where $\dot{\tilde{\rho}}_\phi$ is the autodiff derivative of the *slow* envelope and $\tilde{\mathcal{L}}_{\Theta, t}$ is the GKSL generator of Theorem 7; for single-transition jump operators $\tilde{\mathcal{L}}_{\Theta, t}$ is time-independent. Because $\tilde{\rho}_\phi$ is slowly varying, $\dot{\tilde{\rho}}_\phi$ is far better conditioned under measurement noise than the lab-frame derivative, which is what makes the dissipative rates recoverable. We therefore add a *two-stage rate readout*: the interaction-picture generator is linear in the rates, $\tilde{\rho} = \sum_k \gamma_k \mathcal{D}_k[\tilde{\rho}]$ with $\mathcal{D}_k[\rho] = L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\}$, so after fitting the envelope we solve the small nonnegative least-squares problem $\min_{\gamma \geq 0} \sum_i \left\| \dot{\tilde{\rho}}_\phi(t_i) - \sum_k \gamma_k \mathcal{D}_k[\tilde{\rho}_\phi(t_i)] \right\|_F^2$ —the operational form of the identifiability bound (Theorem 5).

Appendix E: Operator-system spectral truncation

The interaction-picture generator $\tilde{\mathcal{L}}_t$ is almost periodic; on $[0, T]$ it admits the Fourier series $\tilde{\mathcal{L}}_t = \sum_k e^{ik\omega_0 t} \mathcal{L}_k$, $\omega_0 = 2\pi/T$, whose modes $\mathcal{L}_k = \frac{1}{T} \int_0^T e^{-ik\omega_0 t} \tilde{\mathcal{L}}_t dt$ we obtain by a temporal FFT of the dense superoperator on a uniform grid. The *level- N spectral truncation* keeps the band $|k| \leq N$, $\tilde{\mathcal{L}}_t^{(N)} = \sum_{|k| \leq N} e^{ik\omega_0 t} \mathcal{L}_k$ —a projection onto the finite operator system spanned by the slow eigen-operators, in the sense of Connes–van Suijlekom [40, 41]. Retaining $\mathcal{L}_0$ and conjugate-symmetric pairs $\mathcal{L}_{\pm k}$ keeps the truncated generator Hermiticity- and trace-preserving; $N = 0$ is the time-independent secular (Davies) generator [43] and $N = 1$ the rotating-wave approximation. The out-of-band weight $\eta_N = \sum_{|k| > N} \|\mathcal{L}_k\|$ is the a posteriori certificate of Theorem 8; the full band

reproduces the structured kernel of Eq. (A1) exactly.

Appendix F: Intermediate representation, dispatch, and algorithms

The compiler receives an IR $\text{IR} = (d, \{G_\ell\}, \{F_a\}, \Theta, \{O_j\}, \text{batch}, \text{mode})$. Dense mode constructs Eq. (A2); structured mode evaluates Eq. (A1) directly and is the default when d is large, m is small, or the computation is batched. A practical dispatch rule is

$$\text{mode} = \begin{cases} \text{dense}, & d \leq d_{\text{dense}}, \text{ small batch,} \\ \text{structured}, & md^2 \ll d^4 \text{ or large batch,} \\ \text{tensor}, & H, L_k \text{ tensor-local.} \end{cases} \quad (\text{F1})$$

Thresholds should be empirical, because BLAS libraries, GPU memory, and batch sizes move the crossover.

Algorithm 1. Residual-kernel dispatch from an open-system IR

Require: IR with dimension d , jump count m , sparsity/tensor metadata, batch size N_b , device.

- 1: **if** tensor-local metadata available and tensor backend enabled **then**
- 2: return tensor-local residual kernel.
- 3: **else if** d^4 storage fits device memory and $d \leq d_{\text{dense}}$ **then**
- 4: materialize dense Liouvillian; return dense matvec kernel.
- 5: **else if** operators are sparse **then**
- 6: return sparse structured kernel.
- 7: **else**
- 8: return dense-matrix structured kernel (commutator and dissipators term by term).
- 9: **end if**
- 10: Attach gradient rule: automatic differentiation, custom adjoint, or finite-difference fallback.

Algorithm 2. Structure-preserving, physics-informed training

Require: Observations $\{(t_i, O_j, y_{ij}, m_{ij})\}$, initial state ρ_0 , Hamiltonian basis $\{G_\ell\}$, jump/Kossakowski basis, collocation grid $\{\tau_\ell\}$.

Ensure: Density surrogate $\rho_\phi(t)$ and generator parameters Θ .

- 1: Initialize ϕ and unconstrained generator parameters ξ ; compile the IR to a residual evaluator.
- 2: **for** step $s = 1, \dots, S$ **do**
- 3: Form $A_\phi(t)$ by Eq. (3) and $\rho_\phi(t)$ by Eq. (2); map rates by Eq. (B1) or $C = BB^\dagger$.
- 4: Evaluate predictions $\text{Tr}[O_j \rho_\phi(t_i)]$, derivative $\dot{\rho}_\phi(\tau_\ell)$, and $\mathcal{L}_\Theta[\rho_\phi(\tau_\ell)]$.
- 5: Compute $\mathcal{J} = \lambda_{\text{data}} \mathcal{J}_{\text{data}} + \lambda_{\text{phys}} \mathcal{J}_{\text{phys}} + \lambda_0 \mathcal{J}_0$; update (ϕ, ξ) with Adam.
- 6: **end for**
- 7: Return (ϕ, Θ) and the residual certificate of Theorem 4.

Algorithm 3. Spectral-frame training and rate readout

Require: Observations $\{(t_i, O_j, y_{ij})\}$, initial state ρ_0 , frame Hamiltonian H , jump operators $\{L_k\}$, collocation grid $\{\tau_\ell\}$.

Ensure: Lab trajectory $\rho_\phi(t)$, fitted rates γ .

- 1: Diagonalize $H = \sum_a E_a |a\rangle\langle a|$; precompute the Bohr phases $e^{i\omega_{ab}t}$ and the eigenbasis jump operators $\tilde{L}_k = U^\dagger L_k U$.
- 2: **for** step $s = 1, \dots, S$ **do**
- 3: Form the envelope $\tilde{\rho}_\phi(t)$ by Eqs. (3)–(2) and the lab state $\rho_\phi(t) = U(t) \tilde{\rho}_\phi(t) U(t)^\dagger$ by Eq. (D1).
- 4: Evaluate $\text{Tr}[O_j \rho_\phi(t_i)]$ for the data term and the interaction-picture residual (D2) for $\mathcal{J}_{\text{phys}}^{\text{spec}}$; update (ϕ, ξ) with Adam.
- 5: **end for**
- 6: **Rate readout:** with the trained envelope, solve $\min_{\gamma \geq 0} \sum_i \|\tilde{\rho}_\phi(t_i) - e^{t_i \sum_k \gamma_k \mathcal{D}_k} \tilde{\rho}_\phi(0)\|_F^2$ (integral form) or the derivative least squares of Appendix D.

7: Return ρ_ϕ and γ .

Algorithm 4. Operator-system spectral truncation of the generator

Require: Frame H , residual coherent term $H_c(t)$, jump operators $\{L_k(t)\}$, window T , level N .

Ensure: Level- N truncated generator and certificate η_N .

- 1: Sample the interaction-picture Liouvillian $\tilde{\mathbb{L}}(t_g)$ on a uniform grid; take a temporal FFT to obtain Fourier modes $\{\mathcal{L}_k\}$.
 - 2: Keep the band $|k| \leq N$ (the $k=0$ mode and conjugate-symmetric pairs); set $\tilde{\mathbb{L}}^{(N)}(t) = \sum_{|k| \leq N} e^{ik\omega_0 t} \mathcal{L}_k$.
 - 3: Compute the certificate $\eta_N = \sum_{|k| > N} \|\mathcal{L}_k\|$ (Theorem 8); propagate $\tilde{\mathbb{L}}^{(N)}$ and rotate the envelope back to the lab frame as needed.
-

Appendix G: Proofs

Proof of Theorem 1. Fix $t \in [0, T]$ and write $A = A(t)$, $M = AA^\dagger$, $Z = \text{Tr } M$, so that $\rho_A(t) = M/Z$. We verify the three defining properties of $\mathcal{D}(\mathcal{H})$ in turn, then differentiability.

(i) *Hermiticity.* Taking the conjugate transpose and using $(XY)^\dagger = Y^\dagger X^\dagger$ and $(A^\dagger)^\dagger = A$ gives $M^\dagger = (AA^\dagger)^\dagger = (A^\dagger)^\dagger A^\dagger = AA^\dagger = M$, so M is Hermitian. Since $Z = \text{Tr } M$ is real (the trace of a Hermitian matrix is the sum of its real diagonal entries), $\rho_A = M/Z$ is Hermitian.

(ii) *Positive semidefiniteness.* For any $v \in \mathbb{C}^d$,

$$v^\dagger M v = v^\dagger A A^\dagger v = (A^\dagger v)^\dagger (A^\dagger v) = \|A^\dagger v\|_2^2 \geq 0,$$

so $M \succeq 0$. We now show $Z > 0$. Because $A \neq 0$ there is an entry $A_{pq} \neq 0$, hence $Z = \text{Tr}(AA^\dagger) = \sum_{i,j} |A_{ij}|^2 = \|A\|_F^2 \geq |A_{pq}|^2 > 0$. Dividing a PSD matrix by a strictly positive scalar preserves positive semidefiniteness, so $\rho_A = M/Z \succeq 0$.

(iii) *Unit trace.* By linearity of the trace and $Z = \text{Tr } M$, $\text{Tr } \rho_A = \text{Tr}(M/Z) = (\text{Tr } M)/Z = Z/Z = 1$.

Properties (i)–(iii) are exactly the conditions defining $\mathcal{D}(\mathcal{H})$, so $\rho_A(t) \in \mathcal{D}(\mathcal{H})$.

Differentiability and tracelessness of the derivative. Suppose A is differentiable on $[0, T]$. Then $M(t) = A(t)A(t)^\dagger$ is differentiable with $\dot{M} = \dot{A}A^\dagger + A\dot{A}^\dagger$ by the product rule, and $Z(t) = \text{Tr } M(t)$ is differentiable with $\dot{Z} = \text{Tr } \dot{M}$. Since $Z(t) > 0$ for all t by (ii), the quotient $\rho_A = M/Z$ is differentiable on $[0, T]$ with

$$\dot{\rho}_A = \frac{\dot{M}}{Z} - \frac{M\dot{Z}}{Z^2}.$$

Each term on the right is Hermitian [$\dot{M}$ is Hermitian because $(\dot{A}A^\dagger + A\dot{A}^\dagger)^\dagger = \dot{A}A^\dagger + A\dot{A}^\dagger$, and $Z, \dot{Z} \in \mathbb{R}$], so $\dot{\rho}_A$ is Hermitian. Finally, differentiating the identity $\text{Tr } \rho_A(t) \equiv 1$ from (iii) and using linearity of the trace gives $\text{Tr } \dot{\rho}_A(t) = \frac{d}{dt} \text{Tr } \rho_A(t) = 0$. This holds at every t , every training iteration, and every evaluation point, which is exactly the zero positivity-violation rate measured for the Cholesky model in Experiment 1 (Table I). $\square$

Proof of Theorem 2. The proof has three steps: (1) regularize the target trajectory so it is uniformly positive

definite; (2) extract a continuous Cholesky factor; (3) approximate that factor by a network and push the approximation through the (locally Lipschitz) density map.

Step 1: regularization. Fix $\eta \in (0, 1)$ and define $\rho_\eta(t) = (1 - \eta)\rho(t) + \eta\sigma$. As a convex combination of the state $\rho(t) \succeq 0$ and $\sigma \succeq 0$ with weights summing to one, $\rho_\eta(t) \in \mathcal{D}(\mathcal{H})$. Moreover for any unit vector v , $v^\dagger \rho_\eta(t) v \geq \eta v^\dagger \sigma v \geq \eta \lambda_{\min}(\sigma) =: c > 0$ because σ is full rank; hence $\rho_\eta(t) \succeq cI$ uniformly in t , i.e. ρ_η is uniformly positive definite. Its deviation from the target is bounded by

$$\begin{aligned} \sup_t \|\rho_\eta(t) - \rho(t)\|_F &= \sup_t \|\eta(\sigma - \rho(t))\|_F \\ &= \eta \sup_t \|\sigma - \rho(t)\|_F =: \eta K, \end{aligned}$$

where $K < \infty$ because ρ is continuous on the compact interval $[0, T]$ and $\mathcal{D}(\mathcal{H})$ is bounded. Choose $\eta = \min\{1/2, \varepsilon/(2K + 1)\}$, so that $\sup_t \|\rho_\eta - \rho\|_F < \varepsilon/2$.

Step 2: continuous Cholesky factor. For each t , $\rho_\eta(t)$ is Hermitian positive definite, so it has a unique Cholesky factorization $\rho_\eta(t) = C_\eta(t)C_\eta(t)^\dagger$ with $C_\eta(t)$ lower triangular and strictly positive real diagonal. The Cholesky factor of a Hermitian positive-definite matrix is a smooth (in particular continuous) function of the matrix entries on the open cone of positive-definite matrices, because each entry of C_η is obtained from entries of ρ_η by the recursive Cholesky formulas, which involve only additions, multiplications, divisions by the strictly positive computed diagonal entries, and square roots of strictly positive computed quantities—all continuous operations on that domain. Since $t \mapsto \rho_\eta(t)$ is continuous and stays in this cone, $t \mapsto C_\eta(t)$ is continuous on $[0, T]$. Collecting the real and imaginary parts of the (at most d^2) free lower-triangular entries gives a continuous map $g : [0, T] \rightarrow \mathbb{R}^{d^2}$.

Step 3: network approximation and Lipschitz transfer. By the universal approximation theorem for feedforward networks with a nonpolynomial activation [38, 39], for every $\varepsilon' > 0$ there is a finite network $\hat{g} : [0, T] \rightarrow \mathbb{R}^{d^2}$ with $\sup_t \|\hat{g}(t) - g(t)\|_2 < \varepsilon'$. Reassembling $\hat{g}(t)$ into a lower-triangular complex matrix $A_\phi(t)$ (the same way g was assembled from C_η) yields a network with $\sup_t \|A_\phi(t) - C_\eta(t)\|_F < \varepsilon'$. Now consider the density map $\Psi(A) = AA^\dagger / \text{Tr}(AA^\dagger)$. On the compact set $\mathcal{K} = \{A : \|A - C_\eta(t)\|_F \leq 1 \text{ for some } t\}$, the trace

$\text{Tr}(AA^\dagger) = \|A\|_F^2$ is bounded below by a constant $\beta > 0$ (because $\inf_t \|C_\eta(t)\|_F > 0$ by uniform positive definiteness and $\varepsilon' \leq 1$), and both the numerator $AA^\dagger$ and the denominator are polynomial in the entries of A with the denominator bounded away from zero; hence Ψ is continuously differentiable on $\mathcal{K}$, so by the mean value inequality it is Lipschitz there with some constant $L < \infty$. Therefore, choosing $\varepsilon' = \min\{1, \varepsilon/(2L)\}$,

$$\begin{aligned} \sup_t \|\Psi(A_\phi(t)) - \Psi(C_\eta(t))\|_F &\leq L \sup_t \|A_\phi(t) - C_\eta(t)\|_F \\ &< L\varepsilon' \leq \varepsilon/2. \end{aligned}$$

But $\Psi(C_\eta(t)) = C_\eta C_\eta^\dagger / \text{Tr}(C_\eta C_\eta^\dagger) = \rho_\eta(t) / \text{Tr} \rho_\eta(t) = \rho_\eta(t)$ since $\text{Tr} \rho_\eta(t) = 1$. Writing $\rho_\phi(t) = \Psi(A_\phi(t))$ and combining with Step 1 via the triangle inequality,

$$\begin{aligned} \sup_t \|\rho_\phi(t) - \rho(t)\|_F &\leq \sup_t \|\rho_\phi(t) - \rho_\eta(t)\|_F \\ &\quad + \sup_t \|\rho_\eta(t) - \rho(t)\|_F < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \end{aligned}$$

and A_ϕ has nonzero output everywhere because $\|A_\phi(t)\|_F \geq \|C_\eta(t)\|_F - \varepsilon' > 0$. This is an *approximation* (existence) guarantee on the hypothesis class; it does not assert that gradient-based optimization will find such a network, and indeed the training objective remains nonconvex—a fragility made explicit by the seed variance of the qutrit reconstruction (Experiment 3, Table III). $\square$

Proof of Theorem 3. We treat the two parameterizations separately and in both cases reduce the learned generator to the canonical GKSL form whose exponential is CPTP by the Gorini–Kossakowski–Sudarshan–Lindblad theorem [1–3], which states that a linear map $\mathcal{L}$ on $d \times d$ matrices generates a completely positive trace-preserving semigroup $\{e^{t\mathcal{L}}\}_{t \geq 0}$ if and only if it can be written as

$$\mathcal{L}[\rho] = -i[H, \rho] + \sum_r \left(V_r \rho V_r^\dagger - \frac{1}{2} \{V_r^\dagger V_r, \rho\} \right), \quad H = H^\dagger. \quad (\text{G1})$$

Rate parameterization. The Hamiltonian $H_\theta = \sum_\ell \theta_\ell G_\ell$ with $G_\ell = G_\ell^\dagger$ and $\theta_\ell \in \mathbb{R}$ is Hermitian because $H_\theta^\dagger = \sum_\ell \theta_\ell G_\ell^\dagger = \sum_\ell \theta_\ell G_\ell = H_\theta$. For each rate, $\text{softplus}(x) = \ln(1 + e^x) > 0$ for all $x \in \mathbb{R}$ (the argument $1 + e^x > 1$, and $\ln$ is positive on $(1, \infty)$); adding $\gamma_{\min} \geq 0$ keeps $\gamma_k(\xi_k) = \text{softplus}(\xi_k) + \gamma_{\min} > 0$. Setting $V_k = \sqrt{\gamma_k} L_k$ absorbs each nonnegative rate into its jump operator, and $V_k \rho V_k^\dagger = \gamma_k L_k \rho L_k^\dagger$, $V_k^\dagger V_k = \gamma_k L_k^\dagger L_k$, so Eq. (A1) is exactly Eq. (G1) with $V_k = \sqrt{\gamma_k} L_k$. Hence $e^{t\mathcal{L}_\theta}$ is CPTP for all $t \geq 0$.

Kossakowski parameterization. Here $\mathcal{L}_\Theta[\rho] = -i[H_\Theta, \rho] + \sum_{a,b=1}^q (C)_{ab} (F_a \rho F_b^\dagger - \frac{1}{2} \{F_b^\dagger F_a, \rho\})$ with $C = BB^\dagger$. The matrix C is Hermitian, since $C^\dagger = (BB^\dagger)^\dagger = BB^\dagger = C$, and PSD, since $u^\dagger C u = u^\dagger B B^\dagger u = \|B^\dagger u\|_2^2 \geq 0$ for all $u \in \mathbb{C}^q$. By the spectral theorem there is a unitary U and nonnegative eigenvalues $\Lambda_r \geq 0$ with $C = U \Lambda U^\dagger$, i.e. $C_{ab} = \sum_r U_{ar} \Lambda_r \bar{U}_{br}$. Define the

rotated operators $\tilde{L}_r = \sqrt{\Lambda_r} \sum_a U_{ar} F_a$. Substituting $C_{ab} = \sum_r U_{ar} \Lambda_r \bar{U}_{br}$ and exchanging the finite sums,

$$\begin{aligned} \sum_{a,b} C_{ab} F_a \rho F_b^\dagger &= \sum_r \Lambda_r \left(\sum_a U_{ar} F_a \right) \rho \left(\sum_b \bar{U}_{br} F_b \right)^\dagger \\ &= \sum_r \tilde{L}_r \rho \tilde{L}_r^\dagger, \end{aligned}$$

where the last equality uses $(\sum_b U_{br} F_b)^\dagger = \sum_b \bar{U}_{br} F_b^\dagger$ and $\Lambda_r = (\sqrt{\Lambda_r})^2$; the identical computation applied to the anticommutator term gives $\sum_{a,b} C_{ab} F_b^\dagger F_a = \sum_r \tilde{L}_r^\dagger \tilde{L}_r$. Therefore $\mathcal{L}_\Theta$ equals Eq. (G1) with $V_r = \tilde{L}_r$ and the same Hermitian H_Θ , so it too generates a CPTP semigroup. Trace preservation holds in both cases because each term of Eq. (G1) is traceless $[\text{Tr}[H, \rho] = 0$ by cyclicity, and $\text{Tr}(V_r \rho V_r^\dagger) = \text{Tr}(V_r^\dagger V_r \rho) = \frac{1}{2} \text{Tr}\{V_r^\dagger V_r, \rho\}$. This architectural guarantee is what makes the learned two-qubit generator a legal Lindbladian that can be propagated to held-out initial states (Experiment 4, Table IV). $\square$

Proof of Theorem 4. Let $\Phi_t = e^{t\mathcal{L}}$ be the CPTP semigroup generated by $\mathcal{L}$ and set the error path $e(t) = \hat{\rho}(t) - \rho(t)$.

Step 1: an inhomogeneous linear ODE for the error. Both $\hat{\rho}$ and ρ are differentiable. Using $\dot{\rho} = \mathcal{L}[\rho]$ and $\dot{\hat{\rho}} = \mathcal{L}[\hat{\rho}] + r$ (the definition of the residual $r = \dot{\hat{\rho}} - \mathcal{L}[\hat{\rho}]$),

$$\dot{e}(t) = \dot{\hat{\rho}}(t) - \dot{\rho}(t) = (\mathcal{L}[\hat{\rho}] + r) - \mathcal{L}[\rho] = \mathcal{L}[e(t)] + r(t),$$

by linearity of $\mathcal{L}$. This is a linear inhomogeneous ODE with constant generator $\mathcal{L}$ and forcing r .

Step 2: variation of constants. The unique solution of $\dot{e} = \mathcal{L}[e] + r$ with initial value $e(0)$ is, by Duhamel's formula (verified by differentiating the right-hand side and using $\frac{d}{dt} \Phi_t = \mathcal{L} \Phi_t = \Phi_t \mathcal{L}$),

$$e(t) = \Phi_t[e(0)] + \int_0^t \Phi_{t-s}[r(s)] ds.$$

Step 3: $r(s)$ is Hermitian and traceless. By hypothesis $\hat{\rho}$ is a Hermitian, trace-one path, so $\dot{\hat{\rho}}$ is Hermitian (the derivative of a Hermitian-valued path is Hermitian) and $\text{Tr} \dot{\hat{\rho}} = \frac{d}{dt} \text{Tr} \hat{\rho} = \frac{d}{dt} 1 = 0$. The generator $\mathcal{L}$ maps Hermitian inputs to Hermitian outputs and is trace annihilating ($\text{Tr} \mathcal{L}[X] = 0$ for all X , since each GKSL term is traceless). Hence $r = \dot{\hat{\rho}} - \mathcal{L}[\hat{\rho}]$ is Hermitian with $\text{Tr} r(s) = 0$.

Step 4: a CPTP map is a trace-norm contraction on Hermitian traceless matrices. Let X be Hermitian with $\text{Tr} X = 0$. By the spectral theorem write the Jordan (positive/negative) decomposition $X = X_+ - X_-$ with $X_\pm \geq 0$ supported on orthogonal eigenspaces; then $\|X\|_1 = \text{Tr} X_+ + \text{Tr} X_-$ and $\text{Tr} X = \text{Tr} X_+ - \text{Tr} X_- = 0$ forces $\text{Tr} X_+ = \text{Tr} X_- =: a$, so $\|X\|_1 = 2a$. If $a = 0$ then $X = 0$ and the claim is trivial; otherwise set $\rho_\pm = X_\pm/a$, which are density matrices, and $X = a(\rho_+ - \rho_-)$. Because Φ is CPTP it maps density matrices to density matrices,

and the trace distance $\frac{1}{2}\|\cdot\|_1$ is contractive under any CPTP map (the Holevo–Helstrom monotonicity of trace distance). Therefore, using linearity and homogeneity of Φ and the norm,

$$\begin{aligned}\frac{1}{2}\|\Phi[X]\|_1 &= \frac{1}{2}\|a(\Phi[\rho_+] - \Phi[\rho_-])\|_1 \\ &= a \cdot \frac{1}{2}\|\Phi[\rho_+] - \Phi[\rho_-]\|_1 \\ &\leq a \cdot \frac{1}{2}\|\rho_+ - \rho_-\|_1 = a = \frac{1}{2}\|X\|_1.\end{aligned}$$

Thus $\|\Phi[X]\|_1 \leq \|X\|_1$ for every Hermitian traceless X , and in particular for each Φ_{t-s} applied to the Hermitian traceless $e(0)$ and $r(s)$.

Step 5: assembling the bound. Take the trace norm of the variation-of-constants formula, apply the triangle inequality (for the integral, $\left\|\int_0^t g \, ds\right\|_1 \leq \int_0^t \|g\|_1 \, ds$), and then the contraction of Step 4 to each Φ :

$$\begin{aligned}\frac{1}{2}\|e(t)\|_1 &\leq \frac{1}{2}\|\Phi_t[e(0)]\|_1 + \frac{1}{2}\int_0^t \|\Phi_{t-s}[r(s)]\|_1 \, ds \\ &\leq \frac{1}{2}\|e(0)\|_1 + \frac{1}{2}\int_0^t \|r(s)\|_1 \, ds,\end{aligned}$$

which is exactly Eq. (5). The right-hand side is computable from the learned path alone (it never references the unknown ρ), so a small training residual is an a posteriori guarantee; the bound is verified empirically on physical perturbations of the exact qubit trajectory in Table VII, where the certificate stays a small constant factor above the true error. $\square$

Proof of Theorem 5. Write $\Delta = \hat{\Theta} - \Theta_* \in \mathbb{R}^q$ and define $\gamma(\alpha) = \Theta_* + \alpha\Delta$ for $\alpha \in [0, 1]$, the straight segment from Θ_* to $\hat{\Theta}$. Since f is continuously differentiable (the observation map is a smooth function of the generator parameters, the solution of a smooth ODE depending smoothly on parameters being smooth), the fundamental theorem of calculus applied to $\alpha \mapsto f(\gamma(\alpha))$ together with the chain rule $\frac{d}{d\alpha}f(\gamma(\alpha)) = J(\gamma(\alpha))\Delta$ gives

$$f(\hat{\Theta}) - f(\Theta_*) = \int_0^1 \frac{d}{d\alpha}f(\gamma(\alpha)) \, d\alpha = \underbrace{\left(\int_0^1 J(\gamma(\alpha)) \, d\alpha\right)}_{=: \bar{J}} \Delta.$$

Lower bound on the smallest singular value of $\bar{J}$. By hypothesis the segment lies in a neighborhood on which the Jacobian varies by at most $s_{\min}/2$, i.e. $\|J(\gamma(\alpha)) - J(\Theta_*)\|_2 \leq s_{\min}/2$ for all $\alpha \in [0, 1]$, where $\|\cdot\|_2$ is the spectral norm. Averaging, $\|\bar{J} - J(\Theta_*)\|_2 = \left\|\int_0^1 (J(\gamma(\alpha)) - J(\Theta_*)) \, d\alpha\right\|_2 \leq \int_0^1 \|J(\gamma(\alpha)) - J(\Theta_*)\|_2 \, d\alpha \leq s_{\min}/2$. The smallest singular value is $s_{\min}(\cdot) = \min_{\|u\|_2=1} \|(\cdot)u\|_2$ and is 1-Lipschitz in the spectral norm by the triangle inequality; since $s_{\min}(J(\Theta_*)) = s_{\min}$,

$$\begin{aligned}s_{\min}(\bar{J}) &\geq s_{\min}(J(\Theta_*)) - \|\bar{J} - J(\Theta_*)\|_2 \\ &\geq s_{\min} - \frac{s_{\min}}{2} = \frac{s_{\min}}{2} > 0.\end{aligned}$$

In particular $\bar{J}$ has full column rank, so $\|\bar{J}\Delta\|_2 \geq s_{\min}(\bar{J})\|\Delta\|_2 \geq \frac{s_{\min}}{2}\|\Delta\|_2$ for every Δ .

Conclusion. Combining with the hypothesis $\|f(\hat{\Theta}) - f(\Theta_*)\|_2 \leq \delta$,

$$\begin{aligned}\delta &\geq \|f(\hat{\Theta}) - f(\Theta_*)\|_2 = \|\bar{J}\Delta\|_2 \geq \frac{s_{\min}}{2}\|\Delta\|_2 \\ \implies \|\hat{\Theta} - \Theta_*\|_2 &\leq \frac{2\delta}{s_{\min}}.\end{aligned}$$

The bound degrades as $s_{\min} \rightarrow 0$, i.e. when the chosen observables do not excite some parameter direction; this is precisely the regime in which the individual qutrit dissipative rates are hard to recover (Experiment 3, Sec. V D), even though physicality is still guaranteed. $\square$

Proof of Theorem 6. We count storage and arithmetic for the two evaluators using the standard model that a product of two dense $d \times d$ matrices costs $\Theta(d^3)$ scalar operations and a dense matrix–vector product with an $n \times n$ matrix costs $\Theta(n^2)$.

Dense Liouvillian. The vectorized generator $\mathbb{L}_\Theta$ of Eq. (A2) acts on $\text{vec}(\rho) \in \mathbb{C}^{d^2}$, hence is a $d^2 \times d^2$ matrix with $d^2 \cdot d^2 = d^4$ entries: storage $O(d^4)$. Constructing it requires, for the Hamiltonian part, the Kronecker products $I \otimes H$ and $H^T \otimes I$, and for each of the m jumps the three Kronecker products $L_k^* \otimes L_k$, $I \otimes L_k^\dagger L_k$, $(L_k^\dagger L_k)^T \otimes I$. Each Kronecker product of two $d \times d$ matrices produces a $d^2 \times d^2$ matrix and costs $\Theta(d^4)$ to form (one scalar multiply per output entry); there are $O(m)$ of them, giving construction cost $O(md^4)$ (the $O(1)$ Hamiltonian Kronecker products are absorbed). Once built, each residual evaluation is the matrix–vector product $\mathbb{L}_\Theta \text{vec}(\rho)$, costing $\Theta((d^2)^2) = \Theta(d^4)$. For N collocation states the dense cost is the one-time construction plus N matvecs: $O(md^4 + Nd^4)$.

Structured (matrix-free) evaluation. Structured mode never forms $\mathbb{L}_\Theta$. Evaluating $\mathcal{L}_\Theta[\rho]$ directly from Eq. (A1) requires: the commutator via $H\rho$ and ρH (two $d \times d$ products); and for each jump k , the products $L_k\rho$, $(L_k\rho)L_k^\dagger$, $C_k := L_k^\dagger L_k$, $C_k\rho$, ρC_k . Each is a $d \times d$ matrix product costing $\Theta(d^3)$, and there is a constant number of them per jump, so one residual evaluation costs $O(md^3)$. Storage is the Hamiltonian H and the m jump operators L_k (and the running ρ and accumulator), i.e. $O(md^2)$. For N collocation states (the C_k can be precomputed once, an $O(md^3)$ term absorbed into the per-state cost), the structured cost is $O(Nmd^3)$.

Comparison. The per-evaluation ratio is $\Theta(d^4)/\Theta(md^3) = \Theta(d/m)$, and the storage ratio is $\Theta(d^4)/\Theta(md^2) = \Theta(d^2/m)$, so for fixed small m the structured route is asymptotically cheaper in both arithmetic and memory as d grows. The robust empirical separation is the *memory* one (Experiment 5, Tables V and VIII): the analytic $O(d^4)$ -versus- $O(md^2)$ storage gap widens with d , and the two evaluators agree to floating-point precision so the structured route is exact, not an approximation. We do not claim the arithmetic

ratio as a wall-clock speedup: our benchmark's dense path rebuilds $\mathbb{L}_\Theta$ on every state (paying the $O(md^4)$ construction once per evaluation rather than amortizing it), so any measured timing gap is dominated by repeated Liouvillian construction and is an implementation artifact, not an intrinsic per-matvec advantage. $\square$

Proof of Theorem 7. Physicality. By hypothesis $\tilde{\rho}(t) \in \mathcal{D}(\mathcal{H})$, i.e. $\tilde{\rho} = \tilde{\rho}^\dagger \succeq 0$ with $\text{Tr } \tilde{\rho} = 1$. For unitary U , $\rho = U\tilde{\rho}U^\dagger$ satisfies $\rho^\dagger = U\tilde{\rho}^\dagger U^\dagger = \rho$; for any v , $v^\dagger \rho v = (U^\dagger v)^\dagger \tilde{\rho} (U^\dagger v) \geq 0$ since $\tilde{\rho} \succeq 0$; and $\text{Tr } \rho = \text{Tr}(U\tilde{\rho}U^\dagger) = \text{Tr}(\tilde{\rho}U^\dagger U) = \text{Tr } \tilde{\rho} = 1$ by cyclicity and $U^\dagger U = I$. Hence $\rho(t) \in \mathcal{D}(\mathcal{H})$ for all t .

Equivalence of the equations of motion. Write $\rho = U\tilde{\rho}U^\dagger$ with $\dot{U} = -iHU$, so $\dot{U}^\dagger = iU^\dagger H$. Differentiating, $\dot{\rho} = -iH\rho + U\dot{\tilde{\rho}}U^\dagger + i\rho H = -i[H, \rho] + U\dot{\tilde{\rho}}U^\dagger$. Substituting $\dot{\rho} = \mathcal{L}_\Theta[\rho] = -i[H_\Theta, \rho] + \mathcal{D}_\Theta[\rho]$ (with $\mathcal{D}_\Theta$ the dissipator) and solving for $\dot{\tilde{\rho}} = U^\dagger(\dot{\rho} + i[H, \rho])U$ gives

$$\begin{aligned} \dot{\tilde{\rho}} &= U^\dagger(-i[H_\Theta - H, \rho] + \mathcal{D}_\Theta[\rho])U \\ &= -i[\tilde{H}_c, \tilde{\rho}] + U^\dagger \mathcal{D}_\Theta[U\tilde{\rho}U^\dagger]U, \end{aligned}$$

using $U^\dagger[X, U\tilde{\rho}U^\dagger]U = [U^\dagger XU, \tilde{\rho}]$ and $\tilde{H}_c = U^\dagger(H_\Theta - H)U$ (Hermitian). For each jump term, $U^\dagger(L_k \rho L_k^\dagger)U = (U^\dagger L_k U)\tilde{\rho}(U^\dagger L_k^\dagger U) = \tilde{L}_k \tilde{\rho} \tilde{L}_k^\dagger$ with $\tilde{L}_k = U^\dagger L_k U$, and likewise $U^\dagger\{L_k^\dagger L_k, \rho\}U = \{\tilde{L}_k^\dagger \tilde{L}_k, \tilde{\rho}\}$ because $U^\dagger L_k^\dagger L_k U = \tilde{L}_k^\dagger \tilde{L}_k$. Thus $\dot{\tilde{\rho}} = \tilde{\mathcal{L}}_{\Theta,t}[\tilde{\rho}]$ with the stated GKSL form, and the chain of identities is reversible, so the two equations are equivalent. Each $\tilde{\mathcal{L}}_{\Theta,t}$ is GKSL because $\tilde{H}_c$ is Hermitian and the jump set $\{\tilde{L}_k(t)\}$ is a valid (unitarily rotated) Lindblad list, hence $e^{s\tilde{\mathcal{L}}_{\Theta,t}}$ is CPTP for frozen t by Theorem 3. This is the architectural physicality and exact equivalence used by the spectral-frame experiments (Experiment 3, Table III). $\square$

Proof of Theorem 8. The map $t \mapsto \tilde{\mathcal{L}}_t$ is a trigonometric/almost-periodic operator-valued function: each matrix element of $\tilde{\mathcal{L}}_t$ is a finite sum of terms $e^{i\Omega t} c$ with Ω a difference of Bohr frequencies (from $\tilde{L}_k(t) = e^{i\omega_{ab}t}(L_k)_{ab}$ and the bilinear structure of the dissipator), so it is bounded and continuous and has a norm-convergent Fourier series on $[0, T]$ with coefficients $\mathcal{L}_k = \frac{1}{T} \int_0^T e^{-ik\omega_0 t} \tilde{\mathcal{L}}_t dt$ and $\sum_k \|\mathcal{L}_k\| < \infty$. For the truncation $\tilde{\mathcal{L}}_t^{(N)} = \sum_{|k| \leq N} e^{ik\omega_0 t} \mathcal{L}_k$, the triangle inequality gives $\sup_t \|\tilde{\mathcal{L}}_t - \tilde{\mathcal{L}}_t^{(N)}\| = \sup_t \|\sum_{|k| > N} e^{ik\omega_0 t} \mathcal{L}_k\| \leq \sum_{|k| > N} \|\mathcal{L}_k\| = \eta_N$, and $\eta_N \rightarrow 0$ as the tail of a convergent series. *Structure preservation:* $\tilde{\mathcal{L}}_t$ maps Hermitian to Hermitian and is trace-annihilating at every t ; equivalently $\mathcal{L}_{-k}[\bar{X}] = \mathcal{L}_k[X]$ and $\text{Tr } \mathcal{L}_k[X] = 0$. Keeping the conjugate-symmetric set $\{0\} \cup \{\pm k : k \leq N\}$ therefore yields a $\tilde{\mathcal{L}}_t^{(N)}$ that is again Hermiticity- and trace-preserving for every t . *Error bound:* let $\tilde{\rho}^{(N)}$ solve $\dot{\tilde{\rho}}^{(N)} = \tilde{\mathcal{L}}_t^{(N)}[\tilde{\rho}^{(N)}]$ with the same initial datum as the exact envelope $\tilde{\rho}$. Treating

$r^{(N)}(t) = \dot{\tilde{\rho}}(t) - \tilde{\mathcal{L}}_t^{(N)}[\tilde{\rho}(t)] = (\tilde{\mathcal{L}}_t - \tilde{\mathcal{L}}_t^{(N)})[\tilde{\rho}(t)]$ as a residual of the truncated dynamics and applying the trace-norm propagation of Theorem 4 (valid because the frozen- t truncated generators are trace-annihilating Hermiticity-preserving maps) bounds the error by the initial error plus $\frac{1}{2} \int_0^t \|r^{(N)}(s)\|_1 ds \leq \frac{1}{2} \int_0^t \eta_N \|\tilde{\rho}(s)\|_1 ds$. At full band $\eta_N \rightarrow 0$, so the truncated model converges to the exact one and the structured kernel is the $N \rightarrow \infty$ special case. The monotone decay of η_N and of the propagated infidelity is measured in Experiment 6 (Fig. 5, Table VI). $\square$

Corollary 15 (Low-rank physicality). *For $A \in \mathbb{C}^{d \times r}$ and $\alpha \geq 0$ not both zero (i.e. $A \neq 0$ or $\alpha > 0$), the matrix $\rho_\phi^{(r)}(t) = (AA^\dagger + \alpha I/d)/(\text{Tr}[AA^\dagger] + \alpha)$ is Hermitian, positive semidefinite, and unit trace. When $\alpha > 0$ the state is full rank; when $\alpha = 0$ its rank is at most r .*

Proof. Let $M = AA^\dagger + \alpha I/d$ and $Z = \text{Tr}[AA^\dagger] + \alpha$. Hermiticity: $(AA^\dagger)^\dagger = AA^\dagger$ and $(\alpha I/d)^\dagger = \alpha I/d$, so $M = M^\dagger$. Positive semidefiniteness: for any $v \in \mathbb{C}^d$, $v^\dagger M v = \|A^\dagger v\|_2^2 + (\alpha/d) \|v\|_2^2 \geq 0$, as a sum of two nonnegative terms; hence $M \succeq 0$. Strict positivity of the denominator: $\text{Tr}[AA^\dagger] = \|A\|_F^2 \geq 0$ and $\alpha \geq 0$, and since not both are zero, $Z = \|A\|_F^2 + \alpha > 0$ (note $\text{Tr}(\alpha I/d) = \alpha$). Thus $Z = \text{Tr } M > 0$ and $\rho_\phi^{(r)} = M/Z \succeq 0$ with $\text{Tr } \rho_\phi^{(r)} = Z/Z = 1$, a density matrix. Rank: $\rho_\phi^{(r)} \succ 0$ (all eigenvalues $\geq \alpha/(dZ) > 0$) iff $\alpha > 0$, so it is full rank then; if $\alpha = 0$, $\rho_\phi^{(r)} = AA^\dagger/Z$ and $\text{rank}(AA^\dagger) \leq \text{rank}(A) \leq r$. This is the variant used for the larger- d structured-compiler regime (Appendix I, Experiment 5). $\square$

Proposition 16 (Frobenius residual certificate via Grönwall). *Let $\mathcal{L}$ be linear with operator norm $\|\mathcal{L}[X]\|_F \leq \Lambda \|X\|_F$ for all X , let ρ solve $\dot{\rho} = \mathcal{L}[\rho]$, and let $\hat{\rho}$ be differentiable with Frobenius residual $\|\dot{\hat{\rho}}(t) - \mathcal{L}[\hat{\rho}(t)]\|_F \leq \delta$ for all t . Then for every $t \geq 0$,*

$$\|\hat{\rho}(t) - \rho(t)\|_F \leq e^{\Lambda t} \|\hat{\rho}(0) - \rho(0)\|_F + \frac{e^{\Lambda t} - 1}{\Lambda} \delta.$$

Proof. Let $e(t) = \hat{\rho}(t) - \rho(t)$ and $r(t) = \dot{\hat{\rho}}(t) - \mathcal{L}[\hat{\rho}(t)]$ with $\|r(t)\|_F \leq \delta$. As in the proof of Theorem 4, $\dot{e} = \mathcal{L}[e] + r$. Let $u(t) = \|e(t)\|_F$. Where $u(t) > 0$ it is differentiable with $\dot{u} = \text{Re} \langle e, \dot{e} \rangle_F / u$, where $\langle X, Y \rangle_F = \text{Tr}(X^\dagger Y)$; by Cauchy-Schwarz and the triangle inequality,

$$\begin{aligned} \dot{u} &= \frac{\text{Re} \langle e, \mathcal{L}[e] + r \rangle_F}{\|e\|_F} \leq \frac{\|e\|_F \|\mathcal{L}[e] + r\|_F}{\|e\|_F} \\ &\leq \|\mathcal{L}[e]\|_F + \|r\|_F \leq \Lambda u + \delta. \end{aligned}$$

(The same differential inequality holds in the Dini/upper-derivative sense at points where $u = 0$, so it is valid for all t .) Multiplying by the integrating factor $e^{-\Lambda t}$ gives

$\frac{d}{dt}(e^{-\Lambda t}u) \leq e^{-\Lambda t}\delta$; integrating from 0 to t ,

$$e^{-\Lambda t}u(t) - u(0) \leq \delta \int_0^t e^{-\Lambda s} ds = \frac{\delta}{\Lambda}(1 - e^{-\Lambda t}),$$

and multiplying back by $e^{\Lambda t}$ yields the claim. The trace-norm certificate of Theorem 4 is usually tighter and more operationally meaningful for quantum states, but this Frobenius form is the quantity directly minimized by the physics loss $\mathcal{J}_{\text{phys}}$, hence easiest to monitor during training. $\square$

Derivative of the Cholesky map. With $M = AA^\dagger$ and $Z = \text{Tr } M$ as in Theorem 1, $\rho = M/Z$ has derivative $\dot{\rho} = \dot{M}/Z - M\dot{Z}/Z^2$ with $\dot{M} = \dot{A}A^\dagger + A\dot{A}^\dagger$ and $\dot{Z} = \text{Tr } \dot{M}$; this $\dot{\rho}$ is Hermitian and traceless (shown in the proof of Theorem 1), and symmetrizing $\dot{\rho} \leftarrow (\dot{\rho} + \dot{\rho}^\dagger)/2$ is a numerical safeguard against floating-point asymmetry before forming the residual.

Appendix H: Proofs of the statistical and approximation theory

This appendix proves the five results of Sec. IV. Throughout, $\|\cdot\|_1$, $\|\cdot\|_2$, and $\|\cdot\|_F$ denote trace, spectral, and Frobenius norms; for a superoperator $\mathcal{G}$ on $d \times d$ matrices, $\|\mathcal{G}\|$ is the operator norm induced by $\|\cdot\|_F$ unless a subscript indicates otherwise, and $\Lambda := \|\mathcal{G}\|$ when $\mathcal{G}$ is understood.

Proof of Proposition 9. (1) *CPTP iff $C \succeq 0$.* By the GKSL theorem in the standard-form (Kossakowski) representation [2, 3], the map $\mathcal{L}_C[\rho] = -i[H, \rho] + \sum_{a,b} C_{ab}(F_a \rho F_b^\dagger - \frac{1}{2}\{F_b^\dagger F_a, \rho\})$ with $H = H^\dagger$ generates a completely positive trace-preserving semigroup if and only if the Kossakowski (GKS) matrix C is Hermitian positive semidefinite. Trace preservation holds for any Hermitian C because each term in the dissipator is trace-annihilating, exactly as in the proof of Theorem 3; the nontrivial content is complete positivity of $e^{t\mathcal{L}_C}$, which by the Gorini–Kossakowski–Sudarshan characterization is equivalent to $C \succeq 0$. Diagonalizing C and using the argument in the Kossakowski part of the proof of Theorem 3, $C \succeq 0$ yields a legal jump list $\{V_r = \sqrt{\Lambda_r} \sum_a U_{ar} F_a\}$ and hence a CPTP semigroup; conversely, a strictly negative eigenvalue $\lambda_r < 0$ of C makes the generator fail the conditional-completeness (Lindblad) test on the maximally entangled state, so $e^{t\mathcal{L}_C}$ is not completely positive for small $t > 0$. Thus $\text{CPTP} \iff C \succeq 0 \iff C_- = 0$.

(2) *Quantitative bound.* Since $\mathcal{L}_C$ is linear in C , the difference generator acts on ρ by

$$(\mathcal{L}_C - \mathcal{L}_C^\sharp)[\rho] = \sum_{a,b} (C_-)_{ab} \left(F_a \rho F_b^\dagger - \frac{1}{2} \{F_b^\dagger F_a, \rho\} \right),$$

because $C = C_+ - C_-$ and the Hamiltonian part cancels. Diagonalize the PSD matrix $C_- = \sum_r |\lambda_r| w_r w_r^\dagger$

(sum over the negative eigenvalues of C , with orthonormal eigenvectors $w_r \in \mathbb{C}^q$) and set $W_r = \sum_a (w_r)_a F_a$. The identical algebra as in the Kossakowski part of the proof of Theorem 3 gives

$$(\mathcal{L}_C - \mathcal{L}_C^\sharp)[\rho] = \sum_r |\lambda_r| \left(W_r \rho W_r^\dagger - \frac{1}{2} \{W_r^\dagger W_r, \rho\} \right).$$

Each dissipator term $\mathcal{D}_W[\rho] = W \rho W^\dagger - \frac{1}{2} \{W^\dagger W, \rho\}$ obeys, by the triangle inequality and submultiplicativity of the Frobenius norm under one-sided multiplication ($\|XY\|_F \leq \|X\|_2 \|Y\|_F$ and $\|XY\|_F \leq \|X\|_F \|Y\|_2$),

$$\|\mathcal{D}_W[\rho]\|_F \leq \|W\|_2^2 \|\rho\|_F + \|W^\dagger W\|_2 \|\rho\|_F = 2\|W\|_2^2 \|\rho\|_F,$$

so $\|\mathcal{D}_W\| \leq 2\|W\|_2^2$. Moreover $\|W_r\|_2 = \|\sum_a (w_r)_a F_a\|_2 \leq \sum_a |(w_r)_a| \|F_a\|_2 \leq \kappa_F \|w_r\|_1 \leq \kappa_F \sqrt{q}$ in general, but the sharper bound used here follows from $\|W_r\|_2^2 \leq \kappa_F^2$ after normalizing the operator basis so that $\sum_a |(w_r)_a|^2 = 1$ acts on orthonormal $\{F_a\}$ (Hilbert–Schmidt orthonormal Hermitian basis, so that $\|W_r\|_2 \leq \kappa_F$ with $\kappa_F = \max_a \|F_a\|_2$). Summing over r and using $\sum_r |\lambda_r| = \|C_-\|_1$,

$$\|\mathcal{L}_C - \mathcal{L}_C^\sharp\| \leq \sum_r |\lambda_r| \|\mathcal{D}_{W_r}\| \leq \sum_r |\lambda_r| \cdot 2\kappa_F^2 = 2\kappa_F^2 \|C_-\|_1,$$

which is Eq. (6). That $\mathcal{L}_C^\sharp$ is a legal GKSL generator is immediate from $C_+ \succeq 0$ and part (1).

(3) *Exact removal.* If $C = BB^\dagger$ then $C \succeq 0$ (for any u , $u^\dagger C u = \|B^\dagger u\|_2^2 \geq 0$), so its Jordan negative part is $C_- = 0$; by part (1) $e^{t\mathcal{L}_C}$ is CPTP for all $t \geq 0$, and the bound (6) is zero, independently of B . Hence the architectural parameterization has identically vanishing CPTP violation. $\square$

Proof of Lemma 10. Contractivity and $\mu \leq 0$. Let $X \in \mathcal{V}_0$, i.e. $X = X^\dagger$ with $\text{Tr } X = 0$. Step 4 of the proof of Theorem 4 shows that any CPTP map Φ satisfies $\|\Phi X\|_1 \leq \|X\|_1$ for such X (the Holevo–Helstrom contractivity of the trace distance on the traceless Hermitian subspace). Applying this to $\Phi = \Phi_h = e^{h\mathcal{L}}$ gives $\|\Phi_h\|_{\mathcal{V}_0} \leq 1$ for all $h \geq 0$. Since $\Phi_h = \text{id} + h\mathcal{L} + o(h)$, the induced operator norm obeys $\|\text{id} + h\mathcal{L}\|_{\mathcal{V}_0} \leq \|\Phi_h\|_{\mathcal{V}_0} + o(h) \leq 1 + o(h)$, so from Eq. (7),

$$\mu(\mathcal{L}) = \lim_{h \downarrow 0} \frac{\|\text{id} + h\mathcal{L}\|_{\mathcal{V}_0} - 1}{h} \leq \lim_{h \downarrow 0} \frac{o(h)}{h} = 0.$$

The exponential bound $\|\Phi_t X\|_1 \leq e^{\mu t} \|X\|_1$ is the standard logarithmic-norm (Coppel) inequality: for $u(t) = \|\Phi_t X\|_1$ one has the right upper Dini derivative $D^+ u(t) = \lim_{h \downarrow 0} h^{-1} (\|(\text{id} + h\mathcal{L})\Phi_t X\|_1 - \|\Phi_t X\|_1) \leq \mu(\mathcal{L}) u(t)$ by definition of the induced logarithmic norm restricted to $\mathcal{V}_0$ (which $\Phi_t X$ belongs to, since Φ_t preserves Hermiticity and trace), and Grönwall’s inequality in Dini form integrates this to $u(t) \leq e^{\mu t} u(0)$.

Strict contraction under primitivity. Suppose Φ_t has a unique full-rank stationary state ρ_∞ and is primitive

with Hilbert–Schmidt spectral gap $g > 0$, i.e. every eigenvalue of $\mathcal{L}$ other than the simple eigenvalue 0 has real part $\leq -g$. Then on the invariant traceless subspace $\mathcal{V}_0$ the semigroup is uniformly asymptotically stable: $\Phi_t|_{\mathcal{V}_0} \rightarrow 0$ as $t \rightarrow \infty$ in any norm (all matrix norms being equivalent in finite dimension), because the restriction $\mathcal{L}|_{\mathcal{V}_0}$ has spectral abscissa $\leq -g < 0$ and hence spectral radius of $\Phi_t|_{\mathcal{V}_0}$ equal to $e^{-gt} < 1$. In particular there is a finite t_* with $\|\Phi_{t_*}\|_{\mathcal{V}_0} =: \vartheta < 1$ (the operator norm falls below one once the transient from non-normality has decayed; such t_* exists because $\|\Phi_t\|_{\mathcal{V}_0} \rightarrow 0$). By the semigroup property and the submultiplicativity of the induced norm, $\|\Phi_{kt_*}\|_{\mathcal{V}_0} \leq \vartheta^k$, so $\mu(\mathcal{L}) \leq t_*^{-1} \ln \vartheta < 0$ by taking $t = kt_* \rightarrow \infty$ in $e^{\mu t} \geq \|\Phi_t\|_{\mathcal{V}_0}$. This proves the strict-contraction claim. $\square$

Proof of Theorem 11. As in Steps 1–3 of the proof of Theorem 4, the error $e(t) = \hat{\rho}(t) - \rho(t)$ is Hermitian and traceless (a difference of trace-one Hermitian paths), obeys the linear inhomogeneous ODE $\dot{e} = \mathcal{L}[e] + r$ with Hermitian traceless residual r , and admits the Duhamel representation $e(t) = \Phi_t[e(0)] + \int_0^t \Phi_{t-s}[r(s)] ds$ with $\Phi_t = e^{t\mathcal{L}}$. Both $e(0)$ and each $r(s)$ lie in $\mathcal{V}_0$. Taking the trace norm, applying the integral triangle inequality, and then the sharp semigroup contraction of Lemma 10 ($\|\Phi_\tau X\|_1 \leq e^{\mu\tau} \|X\|_1$ for $X \in \mathcal{V}_0$) to each factor,

$$\frac{1}{2} \|e(t)\|_1 \leq \frac{1}{2} e^{\mu t} \|e(0)\|_1 + \frac{1}{2} \int_0^t e^{\mu(t-s)} \|r(s)\|_1 ds,$$

which is Eq. (8). If $\mu < 0$ and $\|r(s)\|_1 \leq \bar{r}$ for all s , then $\int_0^t e^{\mu(t-s)} ds = (1 - e^{\mu t})/|\mu| \leq 1/|\mu|$ and $e^{\mu t} \leq 1$, so the right-hand side is bounded by $\frac{1}{2} \|e(0)\|_1 + \bar{r}/(2|\mu|)$ uniformly in $t \geq 0$. Setting $\mu = 0$ recovers Eq. (5) exactly, so Theorem 11 refines Theorem 4. $\square$

Proof of Theorem 12. The proof has three parts: (A) convert the population residual risk into a trajectory error bound via the sharpened stability of Theorem 11; (B) control the gap between empirical and population residual risk by a uniform concentration (Rademacher/covering) argument; (C) assemble and solve for N_c .

(A) *From residual to error.* Set $\mathcal{G} = \tilde{\mathcal{L}}^{(N)}$ and $\mu = \mu(\mathcal{G}) \leq 0$ (Lemma 10; frozen- t GKSL truncated generators are trace-annihilating Hermiticity-preserving, and we use the worst-case constant over $t \in [0, T]$). The surrogate residual is $r(t) = \dot{\rho}_\phi(t) - \mathcal{G}[\rho_\phi(t)]$, and by matched initial condition $e(0) = 0$. Theorem 11 gives

$$\sup_{t \leq T} \frac{1}{2} \|\rho_\phi(t) - \rho^{(N)}(t)\|_1 \leq \frac{1}{2} \int_0^T e^{\mu(T-s)} \|r(s)\|_1 ds \leq \frac{1}{2} \Xi(\mu, T) \sup_{s \leq T} \|r(s)\|_1 \quad (\text{H1})$$

with $\Xi(\mu, T) = \int_0^T e^{\mu u} du = \min\{T, (1 - e^{-|\mu|T})/|\mu|\} \leq \min\{T, 1/|\mu|\}$. We control $\sup_s \|r(s)\|_1$ in L^2 average by the population residual risk: since $\|r(s)\|_1 \leq \sqrt{d} \|r(s)\|_F$ and, more usefully, $\frac{1}{T} \int_0^T \|r(s)\|_F^2 ds = \mathcal{R}(\rho_\phi; \mathcal{G})$ by Definition 1, Jensen’s inequality gives $\frac{1}{T} \int_0^T \|r(s)\|_F ds \leq$

$\mathcal{R}(\rho_\phi; \mathcal{G})^{1/2}$. Using $\|r\|_1 \leq \sqrt{d} \|r\|_F$ and absorbing the dimensional constant into the definition of the norms (or, keeping the Frobenius surrogate error which is what $\mathcal{J}_{\text{phys}}$ minimizes), the time-averaged form of (H1) reads

$$\sup_{t \leq T} \frac{1}{2} \|\rho_\phi(t) - \rho^{(N)}(t)\|_1 \leq \frac{\sqrt{T}}{2} \Xi(\mu, T) \mathcal{R}(\rho_\phi; \mathcal{G})^{1/2}, \quad (\text{H2})$$

where the $\sqrt{T}$ converts the time average of $\|r\|_F$ into the L^2 norm via Cauchy–Schwarz, $\int_0^T \|r\|_F ds \leq \sqrt{T} (\int_0^T \|r\|_F^2 ds)^{1/2} = T^{1/2} \cdot \sqrt{T} \mathcal{R}^{1/2}$ combined with the sup being dominated by the accumulated integral in (H1). Thus the trajectory error is bounded by the square root of the *population residual risk*, up to the dissipative factor.

(B) *Empirical-to-population uniform bound.* The nodes $\tau_1, \dots, \tau_{N_c}$ are i.i.d. uniform on $[0, T]$, and $\hat{\mathcal{R}}(\rho_\phi; \mathcal{G})$ is the empirical mean of the bounded loss $\ell_{\rho_\phi}(\tau) = \|\dot{\rho}_\phi(\tau) - \mathcal{G}[\rho_\phi(\tau)]\|_F^2 \in [0, R_{\max}]$, whose expectation over $\tau \sim \text{Unif}[0, T]$ is $\mathcal{R}(\rho_\phi; \mathcal{G})$. The boundedness $R_{\max} = (G + \Lambda)^2$ follows from Assumption 1: $\|\dot{\rho}_\phi\|_F \leq G$ and $\|\mathcal{G}[\rho_\phi]\|_F \leq \Lambda \|\rho_\phi\|_F \leq \Lambda$. Consider the loss class $\mathcal{S} = \{\tau \mapsto \ell_{\rho_\phi}(\tau) : \rho_\phi \in \mathcal{F}\}$. By the standard symmetrization and Rademacher bound for a uniformly bounded loss class, with probability at least $1 - \delta$, simultaneously over $\rho_\phi \in \mathcal{F}$,

$$|\mathcal{R}(\rho_\phi; \mathcal{G}) - \hat{\mathcal{R}}(\rho_\phi; \mathcal{G})| \leq 2 \mathfrak{R}_{N_c}(\mathcal{S}) + R_{\max} \sqrt{\frac{\log(2/\delta)}{2N_c}}, \quad (\text{H3})$$

where $\mathfrak{R}_{N_c}(\mathcal{S})$ is the empirical Rademacher complexity of $\mathcal{S}$. The loss ℓ is the square of $\tau \mapsto \|\dot{\rho}_\phi(\tau) - \mathcal{G}[\rho_\phi(\tau)]\|_F$, a $\sqrt{R_{\max}}$ -bounded, 1-Lipschitz (in its argument) composition; by Talagrand’s contraction lemma the Rademacher complexity of $\mathcal{S}$ is at most $2\sqrt{R_{\max}}$ times that of the residual-map class $\{\tau \mapsto \dot{\rho}_\phi(\tau) - \mathcal{G}[\rho_\phi(\tau)]\}$. Under Assumption 1 the latter class is parameterized by finitely many bounded weights with covering number $\log \mathcal{N}(\varsigma) \leq \mathfrak{C} \log(1/\varsigma)$, so by Dudley’s entropy integral $\mathfrak{R}_{N_c} \leq c' \sqrt{R_{\max} \mathfrak{C} \log N_c/N_c}$ for a universal c' . Substituting into (H3) and combining the two $N_c^{-1/2}$ terms yields

$$|\mathcal{R} - \hat{\mathcal{R}}| \leq \frac{c R_{\max}}{\sqrt{N_c}} \left(\sqrt{\mathfrak{C} \log N_c} + \sqrt{\log(1/\delta)} \right) =: \varepsilon_{\text{gen}}(N_c, \delta)^2.$$

Hence $\mathcal{R}(\rho_\phi; \mathcal{G}) \leq \hat{\mathcal{R}}(\rho_\phi; \mathcal{G}) + \varepsilon_{\text{gen}}^2 \leq \hat{\mathcal{R}}_\star + \varepsilon_{\text{gen}}^2$, and $\mathcal{R}^{1/2} \leq \hat{\mathcal{R}}_\star^{1/2} + \varepsilon_{\text{gen}}$ by subadditivity of the square root.

(C) *Assembly and complexity.* Substituting $\mathcal{R}^{1/2} \leq \hat{\mathcal{R}}_\star^{1/2} + \varepsilon_{\text{gen}}$ into (H2) gives Eq. (10) with (11). The approximation floor $\hat{\mathcal{R}}_\star^{1/2} = \epsilon_{\text{approx}}$ is attainable because Theorem 2 provides a network whose density path and (by the same construction applied to the C^1 Cholesky factor) time derivative approximate the exact truncated solution uniformly, making the pointwise residual arbitrarily small; the trained empirical risk cannot exceed this floor by more than the optimization gap, which we

absorb into $\widehat{\mathcal{R}}_\star$. To achieve total bracket $\widehat{\mathcal{R}}_\star^{1/2} + \varepsilon_{\text{gen}} \leq \epsilon$ it suffices that $\varepsilon_{\text{gen}} \leq \epsilon - \epsilon_{\text{approx}}$, i.e. $\varepsilon_{\text{gen}}^2 \leq (\epsilon - \epsilon_{\text{approx}})^2$; from (11) this holds once

$$\frac{c R_{\max}}{\sqrt{N_c}} (\sqrt{\mathfrak{C} \log N_c} + \sqrt{\log(1/\delta)}) \leq (\epsilon - \epsilon_{\text{approx}})^2,$$

which, solving for N_c and hiding the $\log N_c$ factor in the $\tilde{O}$, gives the collocation complexity (12). $\square$

Proof of Corollary 13. Error split. By the triangle inequality for the trace distance,

$$\frac{1}{2} \|\rho_\phi(t) - \rho(t)\|_1 \leq \frac{1}{2} \|\rho_\phi(t) - \rho^{(N)}(t)\|_1 + \frac{1}{2} \|\rho^{(N)}(t) - \rho(t)\|_1 \xrightarrow{1} 1 \text{ on any sequence of balls of radius } M_n n^{-1/2} \text{ with } M_n \rightarrow \infty, \text{ because a fixed Gaussian has bounded quantiles, so } \Pi(\|\Theta - \Theta_\star\|_2 > M_n n^{-1/2} \mid y) \rightarrow 0 \text{ in probability (using also } \|\widehat{\Theta}_n - \Theta_\star\|_2 = O_P(n^{-1/2})).$$

The first term is the surrogate error, bounded by the right side of Eq. (10), giving the “surrogate” contribution in Eq. (13) after using $\Xi(\mu, T)$ and $\mathcal{R}^{1/2} \leq \epsilon_{\text{approx}} + \varepsilon_{\text{gen}}$. The second term is the truncation error of Theorem 8: with $\rho^{(N)}$ and ρ sharing the initial datum, Theorem 11 applied to the truncated residual $r^{(N)}(s) = (\widehat{\mathcal{L}}_s - \widehat{\mathcal{L}}_s^{(N)})[\widetilde{\rho}(s)]$ (rotated to the lab frame by the isometry $\rho = U\widehat{\rho}U^\dagger$, which preserves the trace norm) gives $\frac{1}{2} \|\rho^{(N)}(t) - \rho(t)\|_1 \leq \frac{1}{2} \int_0^t e^{\mu(t-s)} \|r^{(N)}(s)\|_1 ds \leq \frac{1}{2} \Xi(\mu, T) \eta_N C_\rho$ with $C_\rho = \sup_s \|\widetilde{\rho}(s)\|_1$, since $\|r^{(N)}(s)\|_1 \leq \eta_N \|\widetilde{\rho}(s)\|_1$ by the operator-norm tail bound of Theorem 8. This is the “truncation” contribution, proving Eq. (13).

Optimal band. At fixed surrogate floor, the only N -dependent term is $\frac{1}{2} \Xi(\mu, T) C_\rho \eta_N$. Under geometric decay $\eta_N \leq \eta_0 \beta^N$, requiring this term to fall to the surrogate-error tolerance ϵ_{tol} demands $\frac{1}{2} \Xi(\mu, T) C_\rho \eta_0 \beta^N \leq \epsilon_{\text{tol}}$, i.e. $\beta^N \leq 2\epsilon_{\text{tol}}/(\Xi C_\rho \eta_0)$; taking logarithms (with $\ln \beta < 0$) and rounding up gives the smallest such integer, Eq. (14) (the factor 2 being absorbed into the tolerance’s definition). Because the per-level cost $\text{cost}(N) = c_0(2N+1)$ is strictly increasing while the certificate is nonincreasing in N and already dominated by the surrogate floor for $N \geq N_\star$, no $N > N_\star$ improves the guaranteed accuracy, so $N_\star$ is simultaneously the minimal-cost band meeting the tolerance and the accuracy-optimal band at bounded budget. $\square$

Proof of Theorem 14. The observation model of Assumption 2 is a regular parametric model: with i.i.d. Gaussian errors of known variance σ^2 , the log-likelihood of the $M = np$ measurements is, up to an additive constant,

$$\ell_M(\Theta) = -\frac{1}{2\sigma^2} \sum_{i,j} (y_{ij} - \text{Tr}[O_j \rho_\Theta(t_i)])^2 = -\frac{1}{2\sigma^2} \|y - f(\Theta)\|_2^2$$

(1)+(2) *Contraction and Bernstein-von Mises.* Because $f \in C^2$ near the interior point $\Theta_\star$ and the Fisher information $\mathcal{I}(\Theta_\star) = \sigma^{-2} J(\Theta_\star)^\top J(\Theta_\star)$ is nonsingular by hypothesis (equivalently $J(\Theta_\star)$ has full column rank, $s_{\min} > 0$, exactly the identifiability condition of Theorem 5), the model satisfies the local asymptotic normality (LAN) conditions: as the number of replications

$M/n \rightarrow \infty$ with n fixed configurations, the normalized score converges to a Gaussian and the likelihood admits the quadratic expansion $\ell_M(\Theta_\star + n^{-1/2}u) = \ell_M(\Theta_\star) + u^\top \Delta_n - \frac{1}{2} u^\top \mathcal{I}(\Theta_\star) u + o_P(1)$ with $\Delta_n \rightsquigarrow \mathcal{N}(0, \mathcal{I}(\Theta_\star))$. The prior π is continuous and positive at $\Theta_\star$. Under these standard regularity conditions the Bernstein-von Mises theorem (see van der Vaart, *Asymptotic Statistics*, Thm. 10.1) applies: the total-variation distance between the posterior law of $\sqrt{n}(\Theta - \widehat{\Theta}_n)$ and $\mathcal{N}(0, \mathcal{I}(\Theta_\star)^{-1})$ (with $\widehat{\Theta}_n$ the MLE, which under LAN is $\sqrt{n}$ -consistent and asymptotically linear) converges to 0 in $P_{\Theta_\star}$ -probability, which is part (2). Contraction (part (1)) is an immediate consequence: the Gaussian limit places posterior mass $\xrightarrow{1} 1$ on any sequence of balls of radius $M_n n^{-1/2}$ with $M_n \rightarrow \infty$, because a fixed Gaussian has bounded quantiles, so $\Pi(\|\Theta - \Theta_\star\|_2 > M_n n^{-1/2} \mid y) \rightarrow 0$ in probability (using also $\|\widehat{\Theta}_n - \Theta_\star\|_2 = O_P(n^{-1/2})$).

(3) *Calibrated coverage.* By part (2), for any Borel set the posterior probability and the limiting Gaussian probability agree asymptotically. Fix $\alpha \in (0, 1)$ and let $\mathcal{C}_{1-\alpha}$ be a level- $(1-\alpha)$ posterior credible set (e.g. the highest-posterior-density region). Its image under the standardizing map converges to the corresponding $\mathcal{N}(0, \mathcal{I}^{-1})$ region, whose frequentist coverage of $\Theta_\star$ is exactly $1-\alpha$ because $\sqrt{n}(\widehat{\Theta}_n - \Theta_\star) \rightsquigarrow \mathcal{N}(0, \mathcal{I}^{-1})$ as well (the MLE and the posterior share the same limiting covariance under BvM); therefore $\Pr(\Theta_\star \in \mathcal{C}_{1-\alpha}) \rightarrow 1-\alpha$. For a fixed prediction target $g(\Theta) = \text{Tr}[O \rho_\Theta(t)]$ with gradient $J_{tO} = \nabla_\Theta g(\Theta_\star)^\top$, the delta method transfers the Gaussian limit: $\sqrt{n}(g(\Theta) - g(\widehat{\Theta}_n))$ has posterior law converging to $\mathcal{N}(0, J_{tO} \mathcal{I}(\Theta_\star)^{-1} J_{tO}^\top)$, so the push-forward credible interval $[g(\widehat{\Theta}_n) \pm z_{1-\alpha/2} \hat{s}_{tO}]$ with $\hat{s}_{tO}^2 = J_{tO} [J^\top J]^{-1} J_{tO}^\top \sigma^2$ has asymptotic coverage $1-\alpha$ for $g(\Theta_\star)$. Its half-width is $z_{1-\alpha/2} \sigma \|J_{tO} [J^\top J]^{-1/2}\|_2$, which is $O(n^{-1/2})$ through the replication scaling of $J^\top J$ and is inflated precisely when J has a small singular value $s_{\min}$ in the direction of J_{tO} —the sloppy-direction phenomenon of Theorem 5. This proves part (3). $\square$

Appendix I: Sensitivity equations and low-rank variants

Differentiating the master equation, $S_a(t) = \partial \rho_\Theta(t) / \partial \Theta_a$ obeys $\dot{S}_a = \mathcal{L}_\Theta[S_a] + (\partial_{\Theta_a} \mathcal{L}_\Theta)[\rho_\Theta]$, $S_a(0) = 0$, and the observation Jacobian of Theorem 5 has entries $J_{(i,j),a} = \text{Tr}[O_j S_a(t_i)]$; the same compiler that evaluates residuals evaluates sensitivities, enabling active experiment design (choosing preparations, observables, and times that maximize the smallest singular value of the sensitivity matrix). For larger d , a rank- r variant uses $A_\phi(t) \in \mathbb{C}^{d \times r}$ with the isotropic floor of Corollary 15, connecting to purification-based simulation.

Appendix J: Experimental implementation details

All experiments are repeated over seeds $\{0, 1, 2, 3, 4\}$ and reported as mean $\pm$ 95% Student- t CI; each seed redraws both the network initialization and the measurement noise. The configuration uses 3000 Adam epochs at learning rate 10^{-3} , 100 time points, and noise standard deviation 0.02. State fidelity is the Uhlmann fidelity computed by a Hermitian eigendecomposition (stable for pure/rank-deficient states).

Single-qubit and sparse-measurement experiments (Secs. VB–VC). The density network (**DensityMatrixNet**) is a four-layer MLP of width 96 (width 64, depth 3 for the sparse study) with GELU activations, mapping t to the entries of a lower-triangular complex Cholesky factor; diagonal entries pass through softplus with a small floor. The soft-penalty and unconstrained baselines (**UnconstrainedDensityNet**) use matched architectures and output the matrix entries directly (the soft-penalty variant adds the eigenvalue-negativity penalty with weight $\lambda_{\text{pos}} = 10$). Loss weights are $\lambda_{\text{data}} = 1$, $\lambda_{\text{phys}} = 0.5$, $\lambda_0 = 10$, with a 60-point (50-point for the sparse study) collocation grid over $t \in [0, 3]$ and the residual derivative obtained by automatic differentiation. The classical least-squares baseline fits $(\omega, \Omega, \gamma_1, \gamma_\phi)$ with `scipy.optimize.least_squares` (trust-region reflective, nonnegativity bounds on rates) against the same noisy observables, propagating the exact Lindblad model at each iteration. The sparse study masks a random fraction of the observation entries.

Qutrit and two-qubit experiments (Secs. VD–VE). The qutrit observations are the three populations together with the real and imaginary coherence quadratures of the (0, 1) and (1, 2) sectors, at 2% noise. We compare three otherwise-identical density networks: a plain MLP, a Fourier-feature network [$K = 48$ deterministic features $\{t, \sin(k\omega_0 t), \cos(k\omega_0 t)\}_{k=1}^K$, $\omega_0 = 2\pi/T$ [44]], and the spectral-frame network (no Fourier features), all width 96, depth 4, trained for the same 3000 Adam epochs. For the spectral-frame network the data term uses the lab state $\rho_\phi = U\tilde{\rho}_\phi U^\dagger$ and the physics residual is the interaction-picture residual (D2); the rates are read out for every method identically by the integral fit of Appendix D applied to the learned interaction-picture envelope (so a method that fails to reconstruct the envelope also fails to identify rates). On the *noiseless* envelope this readout is exact, isolating the residual qutrit rate error to measurement noise. For the two-qubit gate the network ($K = 32$ Fourier features for the residual detuning) is trained on $|00\rangle$; the learned generator (known detunings, learned coupling and rates) is read out and held-out initial states are propagated with the dense Lindblad solver.

Operator-system spectral truncation (Sec. VG). A driven dissipative two-qubit system—detunings $\Delta_1 = 2\pi(5.0)$, $\Delta_2 = 2\pi(4.8)$, a Gaussian-envelope exchange coupling $J(t)(XX+YY)$ with $J_0 = 2\pi(0.4)$, and local amplitude damping $\gamma_1 = 0.1$, $\gamma_2 = 0.08$ —is moved into

TABLE III. **Fast qutrit (leakage) reconstruction (Experiment 3).** Mean state fidelity and mean relative recovery error of the dominant decay rates $(\gamma_{10}, \gamma_{21})$ for a plain MLP, a Fourier-feature density network, and the spectral-frame (interaction-picture) density network. Rates are read out by an integral least-squares fit of the linear-in-rates generator to the learned interaction-picture envelope; this readout is meaningful only for a reconstructed trajectory, so it is reported only for methods reaching mean fidelity ≥ 0.9 (“—” otherwise). Mean $\pm$ 95% CI over 5 seeds.

Density network	Mean state fidelity	Dominant-rate error
Plain MLP	0.7657 ± 0.3463	—
Fourier time-features	0.5238 ± 0.4897	—
Spectral (ours)	0.9847 ± 0.0015	0.273 ± 0.025

the interaction picture set by the detunings. The dense interaction-picture Liouvillian is sampled on a uniform grid (FFT length 4096) over $t \in [0, 3]$, its Fourier modes are truncated to the band $|k| \leq N$, and the level- N generator is propagated with an adaptive RK solver and rotated back to the lab frame; we report the mean and worst-case fidelity to the exact trajectory and the out-of-band weight η_N (Algorithm 4).

Compiler benchmark (Sec. VF). Dense mode uses the Kronecker representation (A2); structured mode evaluates commutator and dissipator terms directly. We report the median residual-evaluation time over repeated runs (warm-up excluded) for a batch of states, and the analytic storage for the dense versus structured representations. The relative error between the two evaluations, $\|\mathcal{L}^{\text{dense}}[\rho] - \mathcal{L}^{\text{struct}}[\rho]\|_F / \max(\|\mathcal{L}^{\text{dense}}[\rho]\|_F, 10^{-15})$, is at floating-point precision.

Appendix K: Extended data

The following tables give the full numerical values summarized graphically in the main text, or that directly verify the theory. Tables III–V report the mean $\pm$ 95% confidence interval over five random seeds for the qutrit reconstruction of Fig. 3(b), the two-qubit gate fidelities of Fig. 3(c), and the compiler benchmark of Fig. 4; Table VI gives the operator-system spectral-truncation hierarchy of Fig. 5. Two further tables transcribe reproduction data not shown in the main figures: Table VII numerically verifies the a posteriori certificate of Theorem 4 on physical perturbations of the exact qubit trajectory, and Table VIII extends the dense-versus-structured benchmark of Theorem 6 to larger Hilbert dimensions (d up to 24), where the asymptotic separation is most visible.

TABLE IV. **Two-qubit dissipative gate (Experiment 4).** State fidelity for the trained $|00\rangle$ trajectory and for three held-out initial states obtained by propagating the learned generator. Mean $\pm$ 95% CI over 5 seeds.

Initial state	Mean fidelity	Final fidelity
$ 00\rangle$ (trained)	0.9944 ± 0.0014	0.9998 ± 0.0001
$ 01\rangle$ (held-out)	0.9504 ± 0.0065	0.9731 ± 0.0092
$ 10\rangle$ (held-out)	0.9551 ± 0.0087	0.9808 ± 0.0086
$ +0\rangle$ (held-out)	0.9743 ± 0.0043	0.9808 ± 0.0045
Average	0.9685 ± 0.0047	—

TABLE V. **Dense-versus-structured compiler scaling (Experiment 5).** Runtime and memory of the two residual evaluators versus Hilbert-space dimension d . Times are the median residual-evaluation wall time over 50 repeats (warm-up excluded) for a batch of 100 states; memory is the analytic storage for the dense $d^2 \times d^2$ Liouvillian versus the structured operators. CPU, single machine. The genuine separation is the analytic memory columns [$O(d^4)$ versus $O(md^2)$]; the time-ratio column is an implementation artifact (the dense path rebuilds the Liouvillian on every state) and is not claimed as an intrinsic speedup—see the main text and Table VIII.

d	Dense (s)	Struct. (s)	Time ratio	Dense (MB)	Struct. (MB)
2	0.0094	0.0015	6.3×	0.000	0.00024
3	0.0130	0.0020	6.5×	0.001	0.00069
4	0.0134	0.0020	6.8×	0.004	0.00122
6	0.0176	0.0021	8.4×	0.020	0.00275
8	0.0291	0.0033	8.8×	0.062	0.00488

Appendix L: Gauge freedom, failure modes, and diagnostics

A Lindblad representation is not unique: unitary rotations of the jump-operator list can leave the dissipator invariant, and for a fixed Kossakowski matrix $C = U\Lambda U^\dagger$ one valid jump set is $\tilde{L}_r = \sum_a U_{ar} \sqrt{\Lambda_r} F_a$, with $U \rightarrow UV$ inside a degenerate eigenspace giving the same generator. Recovery of *individual* jump operators is therefore meaningful only within a fixed basis; the identifiable object is otherwise the generator or channel family. This is why the experiments fix the jump operators and learn a small number of rates. Three diagnostics must be kept separate: state physicality (guaranteed by the Cholesky map), residual size, and held-out prediction error. Common failure modes are residual under-resolution (collocation missing fast features; mitigated by adaptive or causal training [20]), rate sloppiness (near-degenerate observables; small singular values of the sensitivity Gramian), boundary ill-conditioning near pure states (mitigated by an isotropic floor), loss imbalance (mitigated by dynamic weighting), and silent dense fallback (the compiler should log memory before materializing a Liouvillian).

TABLE VI. **Operator-system spectral truncation of the generator (Experiment 6).** Mean and worst-case state fidelity of the level- N spectrally truncated interaction-picture generator propagated against the exact trajectory of a driven dissipative two-qubit system, and the out-of-band spectral weight $\eta_N = \sum_{|k|>N} \|\mathcal{L}_k\|$ that bounds the generator-approximation error (Theorem 8). $N=0$ is the secular (Davies) generator and $N=1$ the rotating-wave approximation; the full band recovers the exact generator (and the structured kernel) to floating-point precision. Deterministic (no seeds).

N	Interpretation	Mean fid.	Max infid.	Out-of-band wt.
0	secular (Davies)	0.759962	5.12e-01	1.660e+01
1	rotating-wave	0.955322	1.11e-01	8.462e+00
2	partial band	0.993096	1.75e-02	3.270e+00
4	partial band	0.999988	3.12e-05	1.819e-01
8	partial band	1.000000	1.15e-08	2.508e-04
16	partial band	1.000000	7.63e-09	2.250e-04

Appendix M: Accelerator-aware implementation

A production implementation should separate the model layer (Hamiltonian terms, jump operators, rate transforms, observables, constraints) from the kernel layer (dense, structured, sparse, or tensor-local execution). The structured residual repeats products involving the same ρ and operators, so a fused implementation should combine anticommutator terms, reuse $C_k = L_k^\dagger L_k$, and keep intermediates in registers/shared memory for small d ; for batches the natural primitive is strided batched GEMM, and for tensor-local terms a sequence of small contractions. Differentiation can use checkpointing, custom vector-Jacobian products for the Lindblad residual, or the adjoint sensitivity equations of Appendix I. Density matrices are sensitive to Hermiticity and trace errors, so mixed precision needs safeguards: symmetrize after residual evaluation, compute traces in higher precision, and monitor minimum eigenvalues on validation grids.

Appendix N: Reproducibility

Every number in this paper is produced by the installable `ctpcpinn` package in the repository's `code/` directory, which exposes the console scripts `ctpcpinn-reproduce` and `ctpcpinn-validate`. In a clean environment, `pip install -e ./code` (or `./code[exact]` for the pinned versions used for the committed numbers) followed by `python code/run_all.py` regenerates all tables and figures into `submission/figures` and `submission/tables` deterministically (≈ 35 min on a laptop CPU); `ctpcpinn-validate` checks the invariants (Hermiticity, positive semidefiniteness, unit trace, and dense=

TABLE VII. **A posteriori residual certificate (Theorem 4).** For a family of physically valid (Hermitian, trace-one) perturbations of the exact single-qubit trajectory of increasing amplitude (column 1, dimensionless), the table reports the maximum trace distance to the exact trajectory actually attained over the time grid (column 2), the certificate bound $\frac{1}{2} \|\hat{\rho}(0) - \rho_0\|_1 + \frac{1}{2} \int_0^t \|r(s)\|_1 ds$ of Eq. (5) computed from the residual alone without knowledge of the exact trajectory (column 3), and their ratio (column 4). The ratio exceeds one at every amplitude, confirming the certificate is a valid upper bound, and stays at a small near-constant factor (≈ 2.35), confirming it is not loose by orders of magnitude. Values are deterministic (single reference trajectory, no seed averaging).

Perturbation amplitude	Max trace distance	Certificate (Thm. 4)	Certificate / error
0.010	0.0039	0.0092	2.37
0.025	0.0097	0.0228	2.35
0.050	0.0193	0.0454	2.35
0.100	0.0387	0.0908	2.35

TABLE VIII. **Extended dense-versus-structured compiler scaling to larger d (Theorem 6).** Analytic storage and measured residual-evaluation time for the dense $d^2 \times d^2$ Liouvillian versus the structured matrix-free kernel, on random Lindblad systems with m jump operators, extended to $d = 24$. Columns: Hilbert dimension d ; number of jump operators m ; dense and structured storage (MB, analytic); dense and structured residual-evaluation time (ms, median over repeated runs on a single CPU); the measured time ratio (dense/structured); and the relative Frobenius difference between the two residual outputs. The genuine, hardware-independent separation is in the analytic memory columns [$O(d^4)$ versus $O(md^2)$], which widen with d , together with the floating-point agreement (relative error $\sim 10^{-16}$, so the structured route is exact, not an approximation). The time ratio (reaching $\approx 277\times$ at $d = 24$) is *not* an intrinsic speedup: in this benchmark the dense path reconstructs the full Liouvillian on every state evaluation, so the ratio is dominated by repeated $O(md^4)$ Liouvillian construction and would largely vanish if the build were amortized across states. We therefore report it only as an implementation artifact. Bold marks the faster evaluator (structured) in each row. Timings are hardware- and implementation-dependent; the analytic memory columns and the floating-point agreement are not.

d	m	Dense (MB)	Structured (MB)	Dense eval (ms)	Structured eval (ms)	Time ratio (artifact)	Relative error
2	2	0.000	0.000	0.090	0.012	$7.55\times$	$3.1e-17$
4	4	0.004	0.001	0.133	0.021	$6.26\times$	$2.1e-16$
8	4	0.066	0.005	0.324	0.026	$12.34\times$	$3.4e-16$
16	4	1.049	0.020	3.340	0.045	$74.83\times$	$6.1e-16$
24	4	5.308	0.046	22.478	0.081	$276.65\times$	$8.1e-16$

structured residual agreement). Each training experiment is repeated over the fixed seeds $\{0, 1, 2, 3, 4\}$ —each redrawing the network initialization and the measurement-noise realization—and reported as mean $\pm$ a 95% Student- t CI; only the wall-clock *times* in Table V are machine dependent (their ratios and the analytic memory columns are not). The package pins BLAS/OpenMP to a single thread internally, which is robust on macOS+conda and costs almost nothing for the small matrices used here ($d = 2\text{--}8$).

The theoretical guarantees map directly onto the software: Theorem 1 (exact physicality) to `models.py:DensityMatrixNet.forward`; Theorem 2 (universal approximation) to `models.py:TimeMLP`; Theorem 3 (GKSL preservation)

to `models.py:PositiveParameter` and the Lindblad right-hand side; Theorem 4 (residual certificate) to `losses.py:physics_residual_loss`; Theorem 5 (local identifiability) to `spectral.py:fit_rates_integral`; Theorem 6 (dense versus structured cost) to `compiler.py` with `lindblad.py:liouvillian_dense`; Theorem 7 (spectral-frame physicality and equivalence) to `models.py:SpectralDensityNet` and `spectral.py:torch_interaction_rhs`; and Theorem 8 (operator-system truncation) to `spectral.py:generator_fourier_modes`, `propagate_truncated`, and `spectral_truncation_error`. A from-first-principles companion narrative, with live pointers to each module, is provided in `ctpcpinn/foundations.py` and `ctpcpinn/theory.py`.

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